

Project HESTIA

C3: COSMIC Capstone Challenge Track 2

Team VT LUNA

05/06/2026

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Advisor: Dr. Kevin Shinpaugh
Mentor: Dr. Jessica Piness



COLLEGE OF ENGINEERING
KEVIN T. CROFTON DEPARTMENT OF
AEROSPACE AND OCEAN ENGINEERING
VIRGINIA TECH

1. Team Introduction

2. Project Introduction

3. Project Overview

4. Regolith Analysis

5. Architecture Analysis

6. Conclusion



Team Organization Chart

PROJECT HESTIA

Advisor



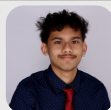
Dr. Kevin
Shinpaugh

Mentor



Dr. Jessica
Piness

Project Lead



Abinav
Khadka

Chief Engineer



Nicholas
Harvey

Microwave/Mold



Ty
Brennan



Nicholas
Harvey



Timmy
Currey

Rover



Juliana
Evans



Jonah
Bradley



Eider
Belda Cano

PTE



Hugh
Young



Robert
Higinbotham



Abinav
Khadka

Filtration Model



Duncan
Foster



Braeden
Peterson

Agenda

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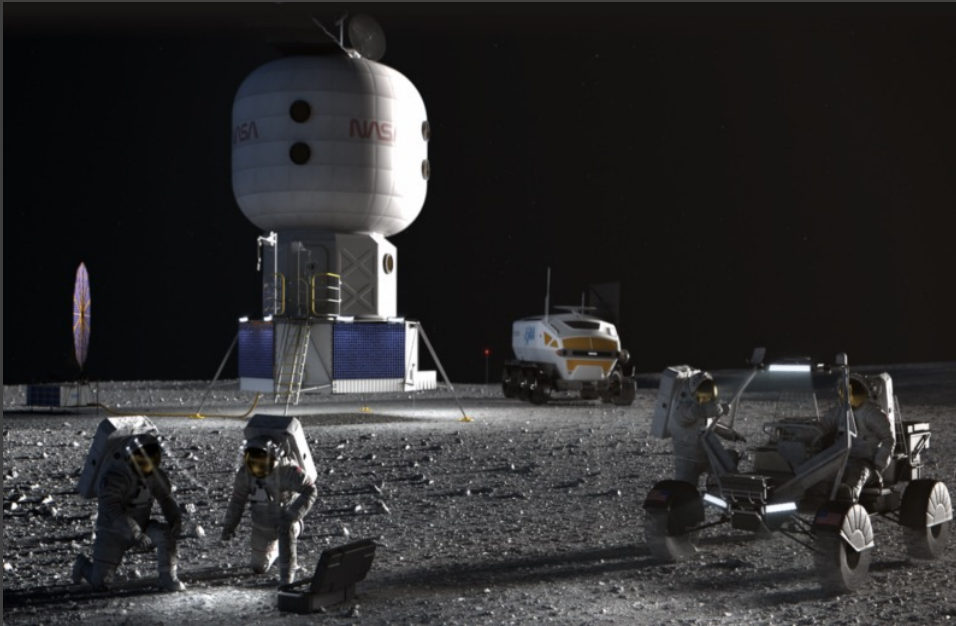


COSMIC Request For Proposal

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C3 COSMIC Track 2 Challenge:

Design a payload for the Griffin lunar lander to help build the infrastructure needed for a permanent lunar outpost



Key Considerations

- What infrastructure is absolutely required?
- What infrastructure will already be in place at the time of our mission?
- How does our design advance lunar surface operations?
- How can we minimize the mass and power requirements of a mission?

Payload is limited to 200kg

Problem Definition

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Problem

Plume ejecta erosion from landing on lunar regolith poses a risk to establishing any permanent lunar infrastructure as high velocity particles can damage landers as well as established habitats and infrastructure.

Proposal

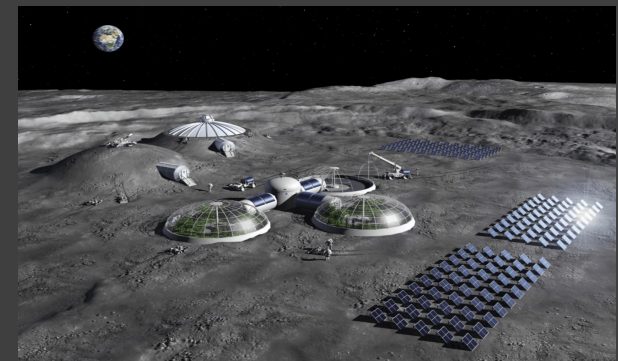
VT Luna proposes designing a payload capable of constructing landing pads on the lunar surface through layered microwave sintering of lunar regolith.

Solution

Construction of lunar landing pads eliminates the threat of plume ejecta and allows for a safe and repeatable landing site, making completion of a lunar colony more feasible.



[2]



[3]

Problem Definition

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Scope

- Produce structurally and thermally sound launch pad tiles can be fabricated repeatedly
- Test the thermal properties of the produced tiles on site
- Recover produced tiles for Earth based testing
- Assemble tiles in a test configuration
- Scalable to produce launchpads via increases production, automation, and terraforming

Timeframe

September 2029: Mission Launch
March 2030: Mission Completion

Assumptions

Griffin Lander

- 5 kW of continuous power during lunar day
- Acts as relay for Earth communications
- Active payload deploy protocol

State of Moon

- No access to water production on the moon
- Local, on-surface astronaut

Illumination

- Nearly constant illumination over 6-month period
- Longest lunar night: 72-96 hours

Astrobotic's Griffin Lunar Lander

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Structure:

- 4.5 m wide by 2 m tall.
- Payload capacity: 625 kg

Power System:

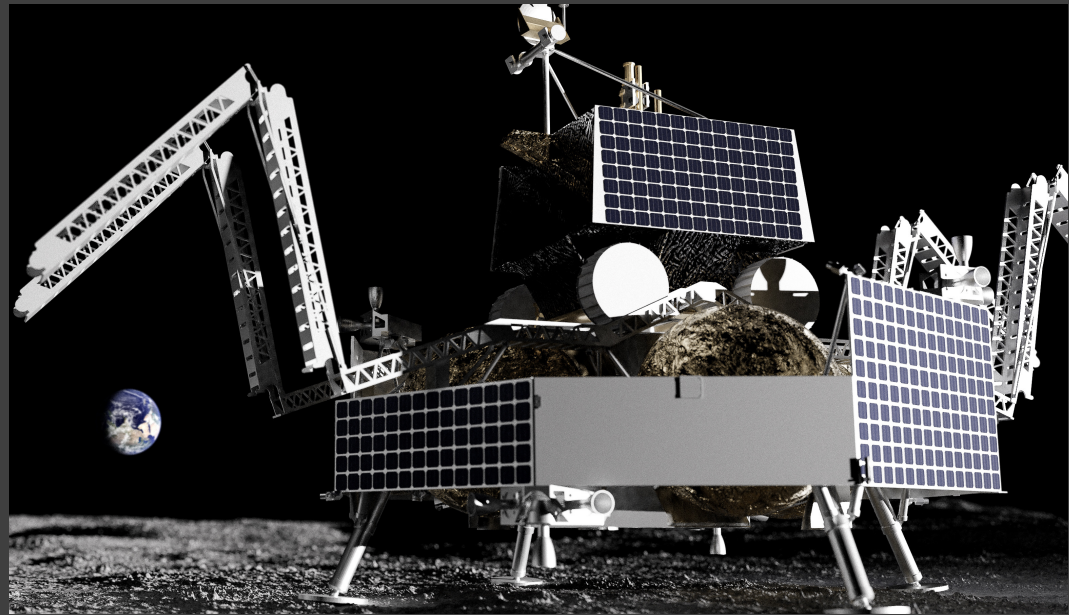
- Solar arrays provide solar energy.
- On board wireless charger.

Communication System:

- 2.4 GHz IEEE 802.11n compliant wireless local area network (WLAN).
- Equipped with a High-Gain Antenna, and Medium-Gain Antenna.

Propulsion System:

- Five 700 lbf pulsed main engines.
- Twelve 25 lbf altitude control system engines.



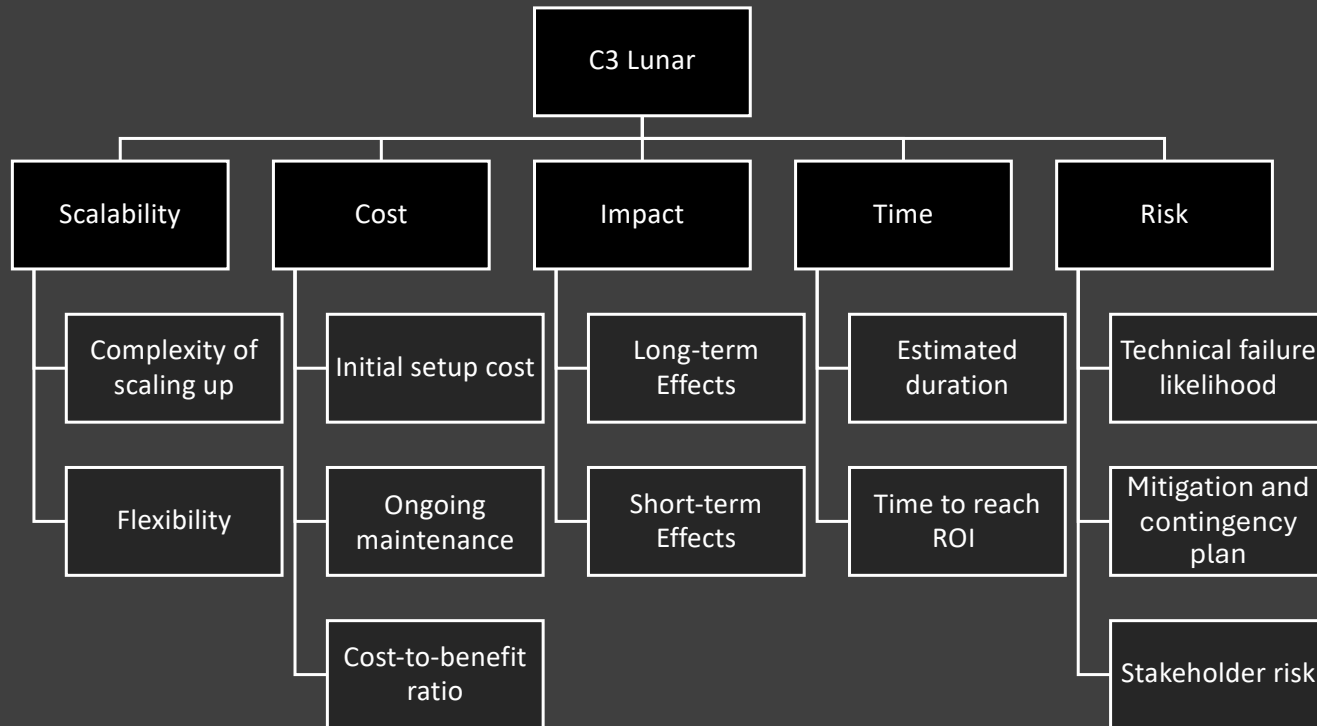
System Requirements

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ID	Requirement Description
RFP1.0	The system shall have a total mass of under 200 kg.
RFP2.0	The system shall occupy a volume no greater than 0.75 of the Griffin Lander payload volume (1.5 m x 1.5 m x 2.5 m)
SYS1.0	The system shall autonomously manufacture structurally sound tiles using In-Situ Resource Utilization (ISRU)
SYS2.0	The system shall accept and process lunar regolith to use as the primary material for tile production.
SYS3.0	The system shall produce tiles that meet defined structural and thermal performance standards.
SYS4.0	The system shall operate autonomously for the duration of the mission, including health monitoring and fault recovery.
SYS5.0	The system shall be capable of surviving and operating within the lunar surface environment (thermal and radiation).
SYS6.0	The system shall perform all operations within the allocated peak power and energy-per-tile budget.
SYS7.0	The system should produce 1 structurally sound tiles autonomously in within 2 weeks.
SYS8.0	The system should detect and respond failures with the Griffin Lander.
SYS9.0	The system should store sufficient regolith to produce a 2-3 tile without additional delivery.

Value System Design

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Factors	Weight	Rank
Scalability	0.444	1
Impact	0.292	2
Time	0.129	3
Risk	0.081	4
Cost	0.052	5

Mass limited to 200kg,
therefore, not considered

Agenda

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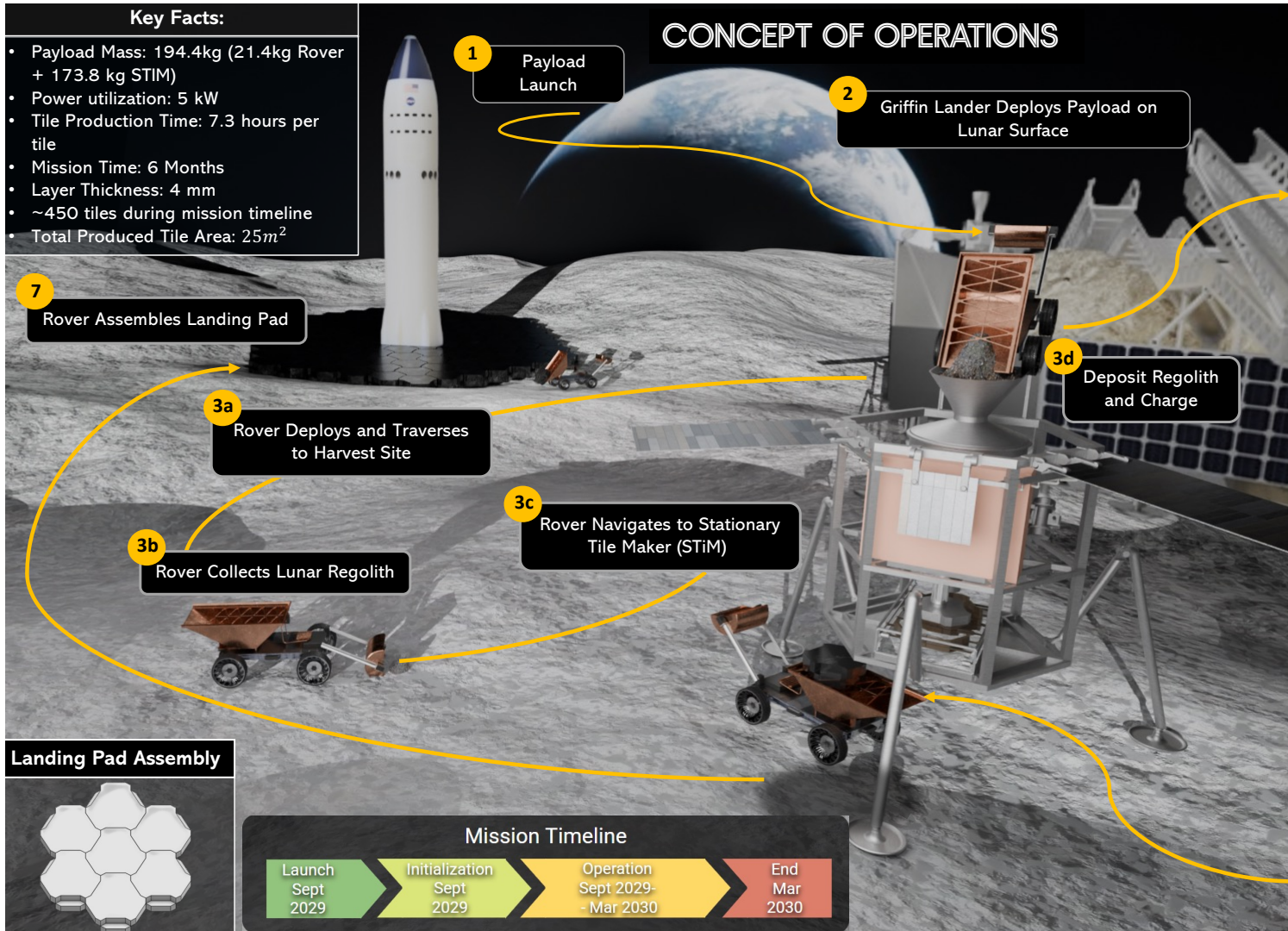
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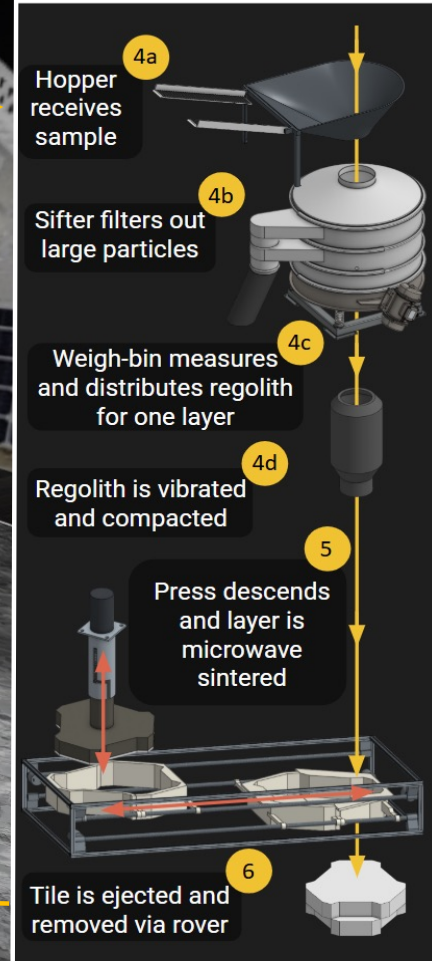
Key Facts:

- Payload Mass: 194.4kg (21.4kg Rover + 173.8 kg STiM)
- Power utilization: 5 kW
- Tile Production Time: 7.3 hours per tile
- Mission Time: 6 Months
- Layer Thickness: 4 mm
- ~450 tiles during mission timeline
- Total Produced Tile Area: $25m^2$

CONCEPT OF OPERATIONS



Regolith Processing and Tile Formation

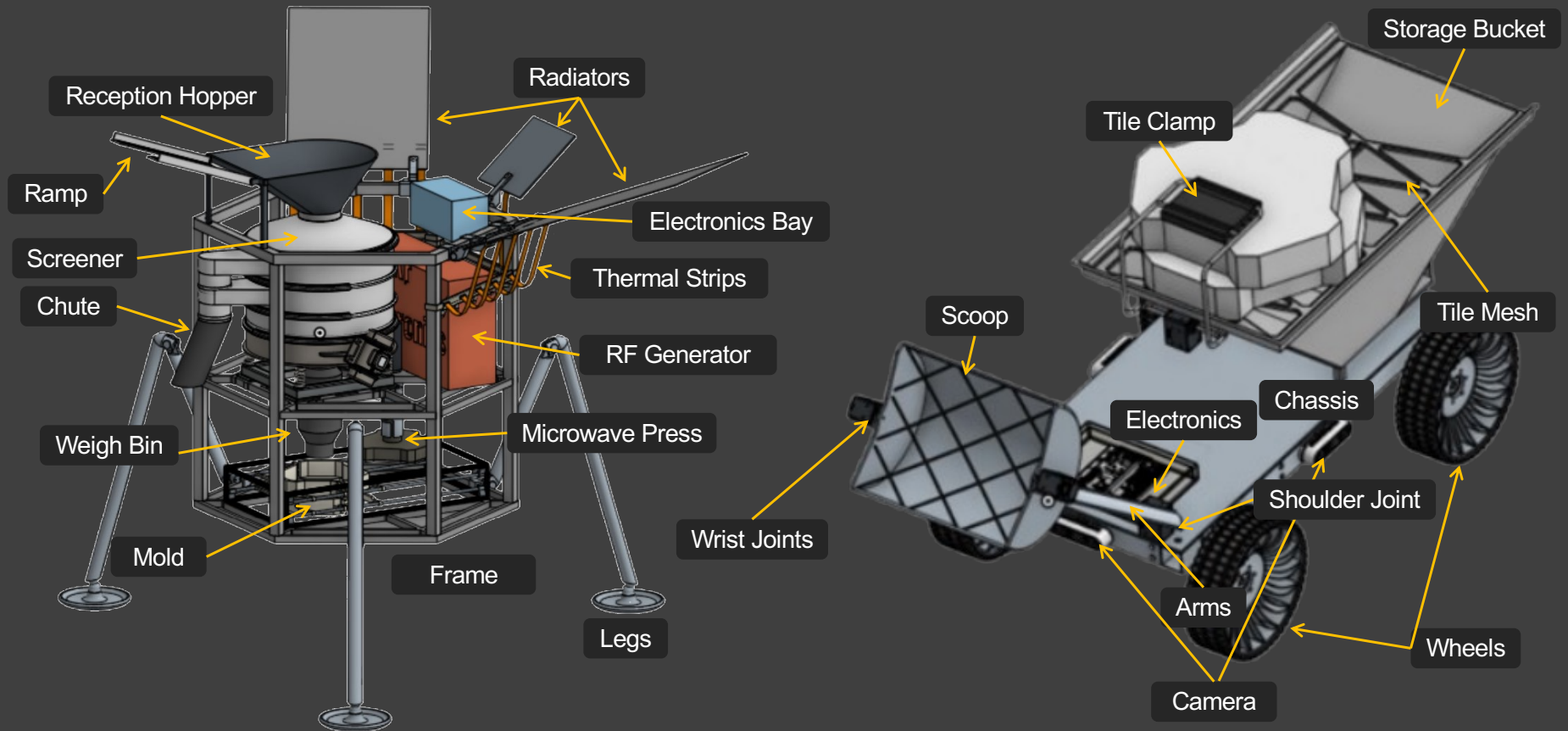


System Overview

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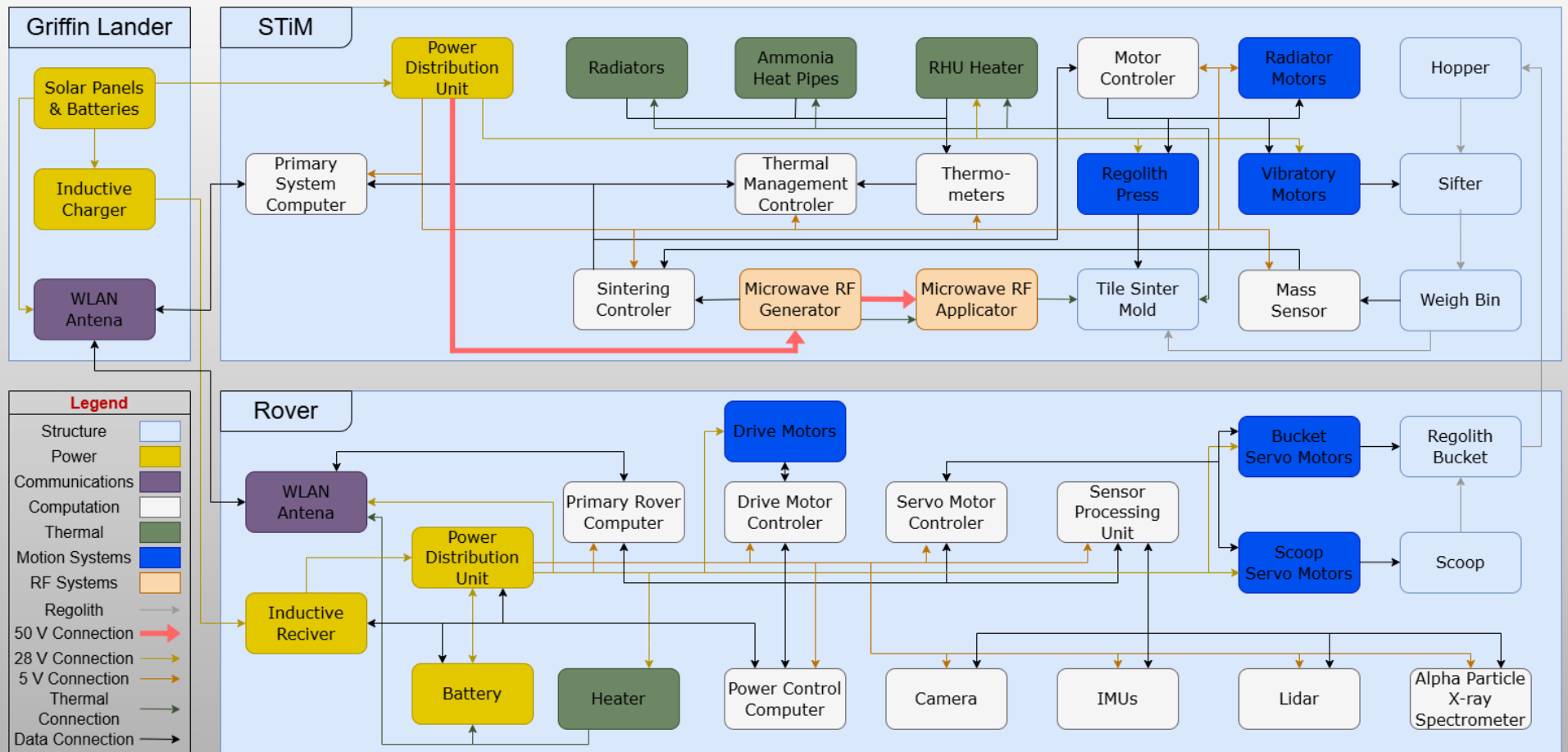
Stationary Tile Maker (STiM)

Rover (Not to scale)



Energy Block Diagram

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Regolith Manufacturing Trade Study

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- Evaluated five regolith processing technologies against mission constraints.
- Microwave Sintering was found to be optimal for the mission due to its scalability and reusability.
- Most other methodologies exceeded our mass constraint

Microwave Sintering	Chemical Binders	Heating Plates	Solar Sintering	Pressed Brick
Reuseable and scalable	Strong chemical bonds	High mass	Little power requirements	Large amount of force generation needed
Uniform heating	Requires use of liquids in space	Slow heating rate	Large tracking optics	Requires some binder
High power requirements	High mass payload to bring binders	High power requirements	High mass and delicate pieces	Contaminates regolith

Technology	Mass(kg)	Power(kW)	Scalability	Tile Strength	Total
Weights	0.30	0.20	0.25	0.25	1
Microwave Sintering	45 ± 10	4	0.45	0.36	0.32
Solar Sinter	95±15	0.15	0.20	0.12	0.22
Chemical Binders	310	0.25	0.10	0.28	0.19
Heating Plates	135±15	5	0.15	0.16	0.16
Pressed Brick	250	2	0.10	0.08	0.11



References: [5-9]

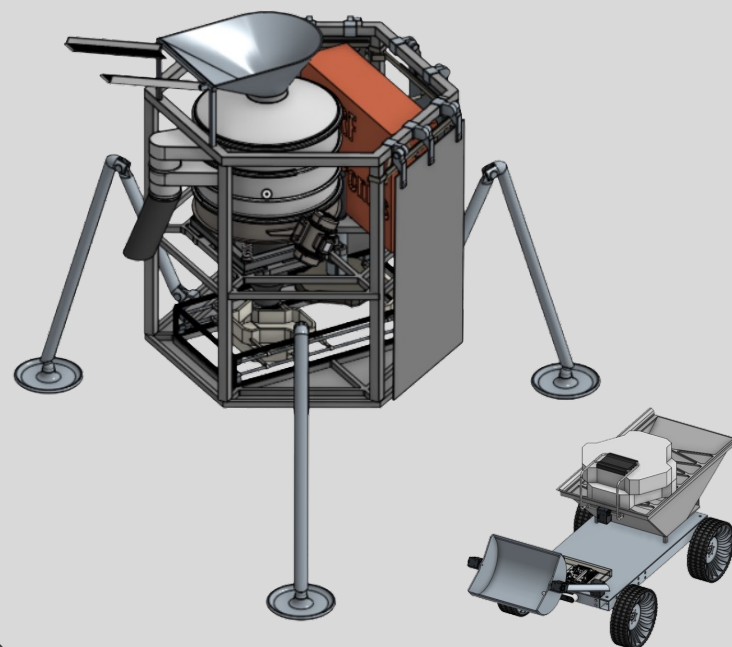
Architecture Selection

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Several Concepts Considered For General Architecture

- **Umbilical Cord**
 - Continuous power via spool-mounted harness
 - Full system capabilities in a single system
 - limited range and mobility complexity when scaled
- **Mobile Manufacturer**
 - Wireless charging, sortie style construction
 - Full system capabilities in a single system
 - Massive battery and reduced operation time
- **STiM and Rover**
 - Division of labor through two interconnected systems
 - Continuous power for the manufacturing plant, smaller battery for the collection system

Architecture Selected: STiM and Rover



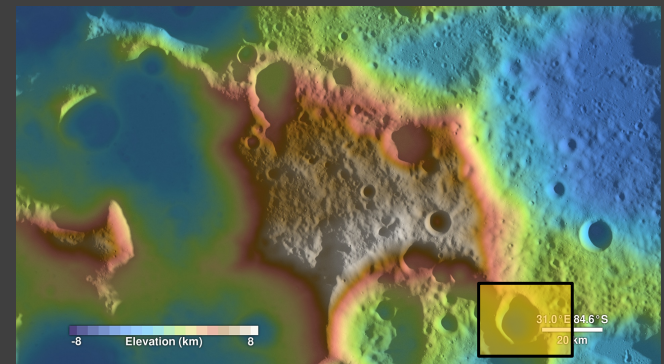
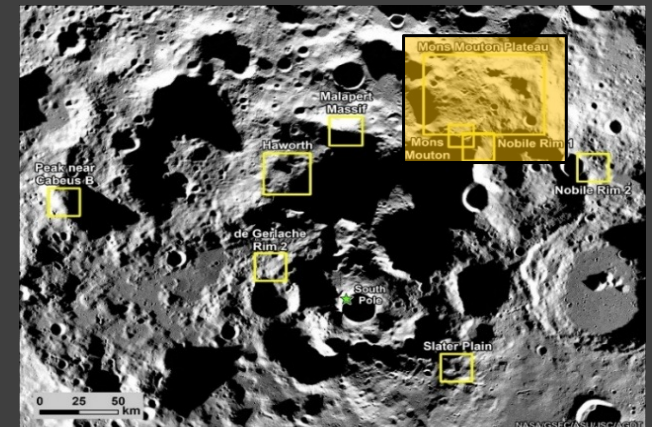
Landing Site Selection

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- Researched NASA's Landing Site Selection for future Artemis lunar missions.
- Evaluated 3 landing site candidates against mission constraints.
- Most important constraint was regolith composition.

	Terrain slope (deg)	Proximity to PSRs score	Illumination (%)	Regolith composition score	Earth visibility (%)	Result
	Minimize		Maximize			
Weights	0.15	0.05	0.25	0.4	0.15	1
Mons Mouton Plateau	7.4 ± 2.3	5	75	4 (Highland)	74 ± 19	4.09
Peak Near Shackleton	~ 7.5	4	85.5	3 (Highland)	~ 50	3.44
De Gerlache rims 2	~ 8	3	88	3 (Highland)	~ 50	3.28

Note: Raw literature values are shown where available; qualitative criteria were converted to scores for the weighted total.
Regolith score based on composition uniformity and suitability for sintering.



Mons Mouton Plateau. An overhead view of Mons Mouton with color-coded elevation.

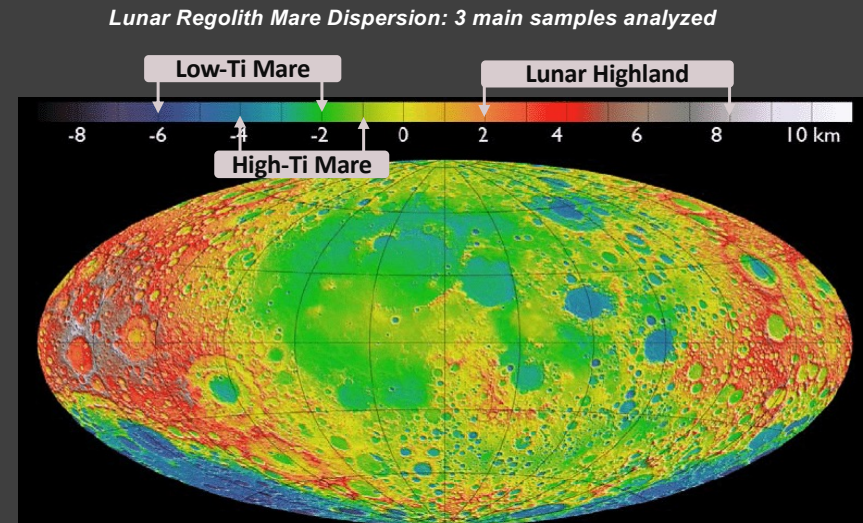
Landing Site Regolith Composition

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Specific Regolith properties we are looking for:

- SiO₂ concentration
- High TiO₂ and FeO concentrations for initial heating
- Low in other contaminants

Chemical Compound	Highland (%)	Low-Ti (%)	High-Ti (%)
Silicon Dioxide	44.9	45.1	41.0
Titanium Dioxide	0.5	2.9	8.5
Aluminum Oxide	25.1	13.1	12.4
Iron Oxide	6.2	17	16.6
Magnesium Oxide	7.5	9.9	8.9
Calcium Oxide	14.9	11.4	11.4



- High-Ti Mare: High-Titanium Mare Regolith
- Low-Ti Mare: Low-Titanium Mare Regolith
- Highland Regolith

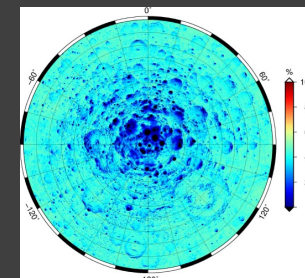
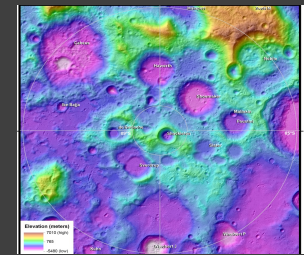
Landing Site Illumination

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- Northeast of Shackelton Crater
- Near constant illumination around the highlands and High Ti Mares at the South Pole

Lunar Morning	Lunar Day	Lunar Evening	Lunar Night
240 Hours	216 Hours	240 Hours	72-96 Hours
5kW	5kW	5kW	0kW

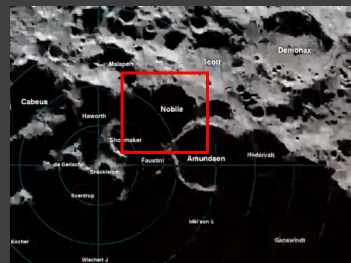
Lunar South Pole receives light at a constant grazing angle, the lander's vertical solar panels are struck at an optimal 90° angle throughout the entire ~26-day lunar day. This ensures peak 5 kW power generation from sunrise to sunset, dropping to 0 kW only during the brief local eclipse where the system enters survival mode.



Lunar Morning



Lunar Day



Lunar Evening



Lunar Night



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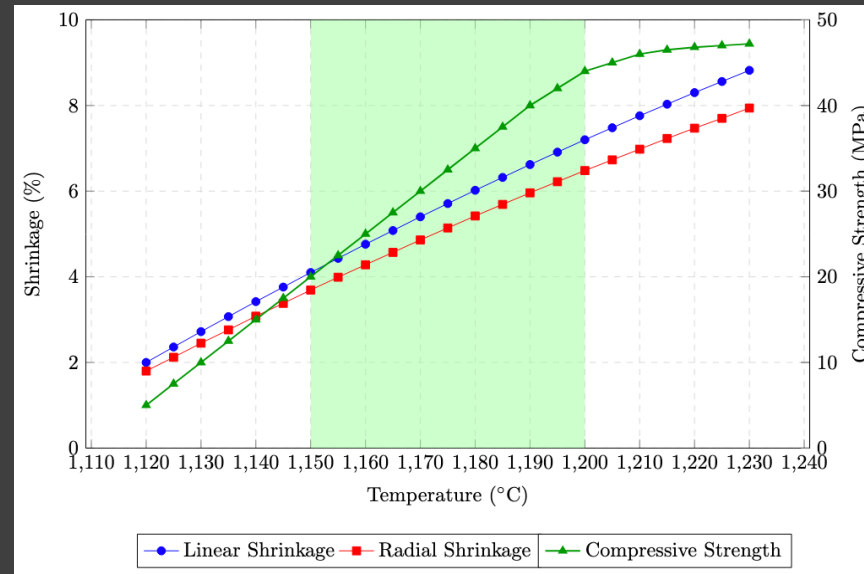


Sintering Properties

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Key Regolith Sintering Properties:

- **Particle Size and Density**
 - Low porosity results in higher compressive strength
 - Average grain size $< 600 \mu\text{m}$ and bulk density $\rho_b = 2.14 \text{ g/cm}^3$ yield the best results
- **Composition:**
 - High in TiO_2 and FeO for initial heating
 - Particle size averaging $< 600 \mu\text{m}$ for sintering
 - Uniform and predictable
- **Mass Loss**
 - 6.5% Mass loss at 1200°C



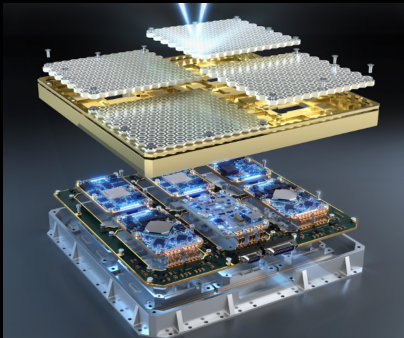
Master material	MM1		MM2		MM3		MM4		MM5		MM6		MM7		MM8	
Specimen code	340	342	345	346	373	374	351	353	356	358	366	367	376	378	368	370
Measured density (g/cm ³)	1.115	1.060	1.120	1.189	1.273	1.280	1.357	1.334	1.368	1.164	0.970	1.387	1.377	1.382	1.762	1.714
Average density for Master Material (g/cm ³)	1.088±0.027		1.155±0.035		1.277±0.003		1.346±0.011		1.266±0.102		1.179±0.209		1.379±0.003		1.738±0.024	

Waveguide Antenna Selection

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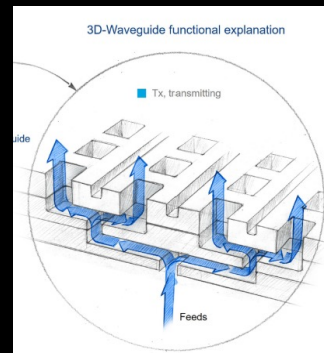
Microwave Sintering Head Requirements
Uniform Tile Shaped 915 MHz antenna
Use the RIM091K5-20 1.5 kW GaN RF Generator

Phased Array



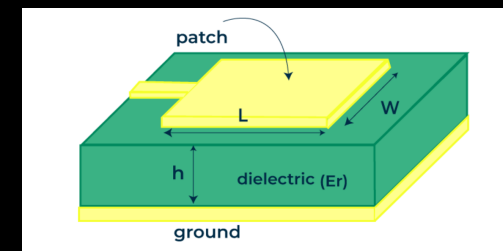
Advanced Beam
Steering and Precise but
Overly Complex

Waveguide



Controlled EM Field
Uniform Heating
Simple and High Power

Patch Antenna

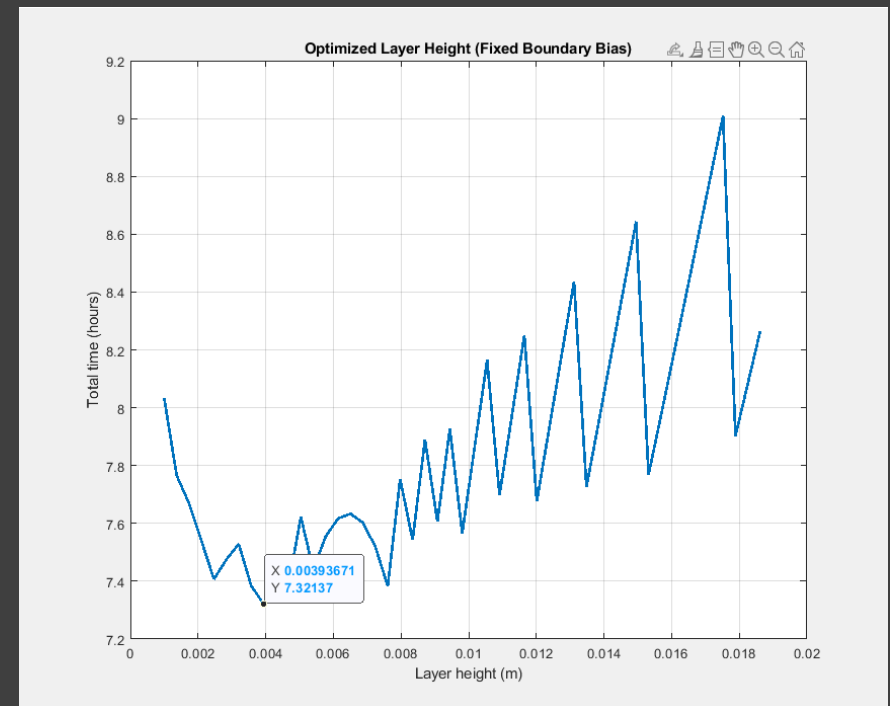


Compact but Localized
EM Field and Uneven
Heating

Microwave Heating Model

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- Simulates regolith heating to calculate tile production time
- Optimizes layer thickness to minimize time
- Calculates
 - Heating time per layer
 - Tile production time
 - Layer thickness
 - Number of layers
- Accounts for
 - Regolith properties
 - Hardware properties
 - Accelerated heating
 - Penetration depth
 - Fill time
 - Conductive heating
 - Radiative losses
- Key Equations
 - $P = KfE^2k\tan(\delta)$
 - $\Delta T = \frac{KfE^2k\tan(\delta)}{\rho C_p}$
- Results
 - Thickness: 3.9mm
 - Number of layers: 27
 - Total time: 7.3 hrs.



<https://www.fst.com/news-stories/technology-innovation/waveguide-antenna/>



References: []

Regolith Heating Simulation

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Takes the exact Chemical Composition of the gathered regolith and can calculate the exact heat time and heating curve under the microwave press

- Allows for an optimized time based on the landing sight chemical makeup
- With variations logged in the system further in-depth studies can categories ideal chemical composition

Alpha Particle X-ray spectrometer will allow for an accurate regolith chemical composition reading

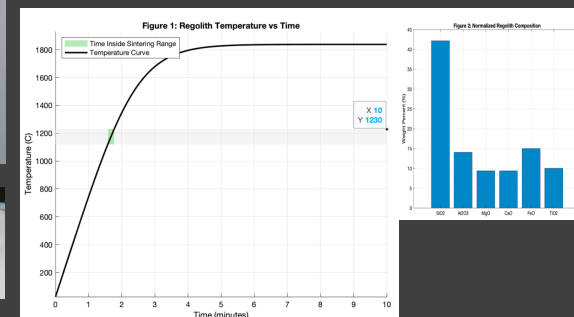
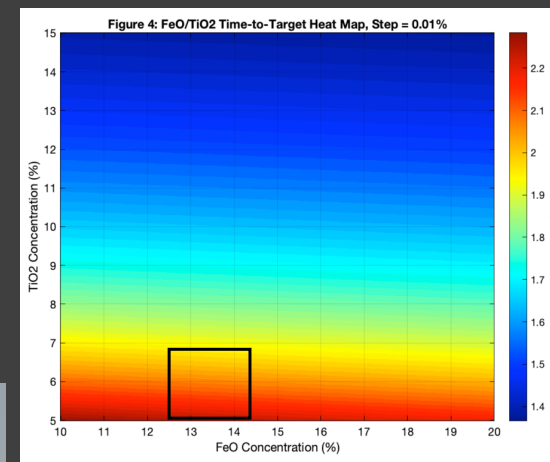
- New miniaturized version allow for readings from the rover

```
Enter oxide concentrations as weight percent.
FeO recommended range: 0 to 20 %
TiO2 recommended range: 0 to 15 %

Enter FeO concentration (%): 15
Enter TiO2 concentration (%): 10
Enter heat map step size (%) [default = 1]: 0.1
Change SiO2/Al2O3/MgO/CaO values? (1=yes, 0=no): 1
SiO2 (%): 45
Al2O3 (%): 15
MgO (%): 10
CaO (%): 10

Calculated Regolith Cp = 766.03 J/kg-K
Best Layer Height: 3.94 mm
Total Time: 7.92 hr

Generating heat map...
Heat map complete.
>>
```

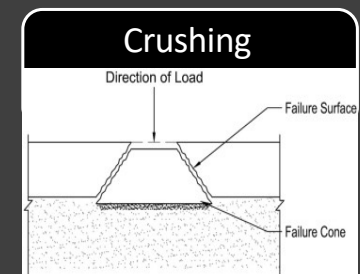
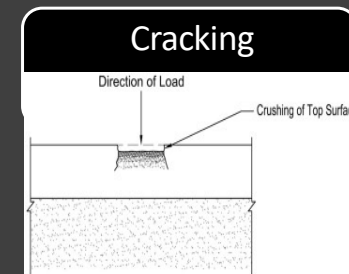
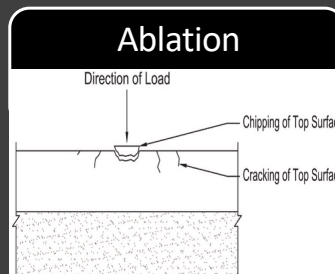
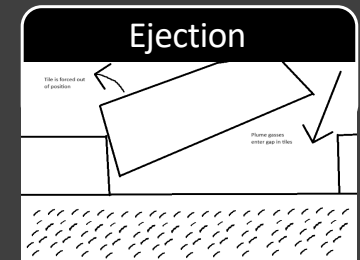
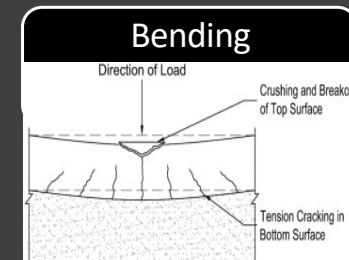
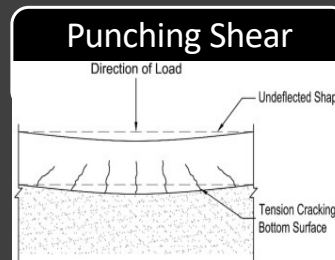


Tile Design Requirements

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ID	Key Tile Requirement	Requirement Justification
TP 1.0	Must interlock securely with limited manipulation.	Ensures tiles are not ejected by the rocket plume
TP 2.0	Must withstand 424 kPa of HLS landing.	Derived from the largest expected lunar lander in Artemis operations, the SpaceX Starship HLS
TP 3.0	Must withstand 26 kPa static pressure of HLS.	Based upon the mass estimates of the Starship HLS
TP 4.0	Must withstand 17800 kW/m ² heating from plume of HLS landing.	Derived from a NASA study simulating a theoretical lunar landings heating characteristics for a variety of vehicle sizes

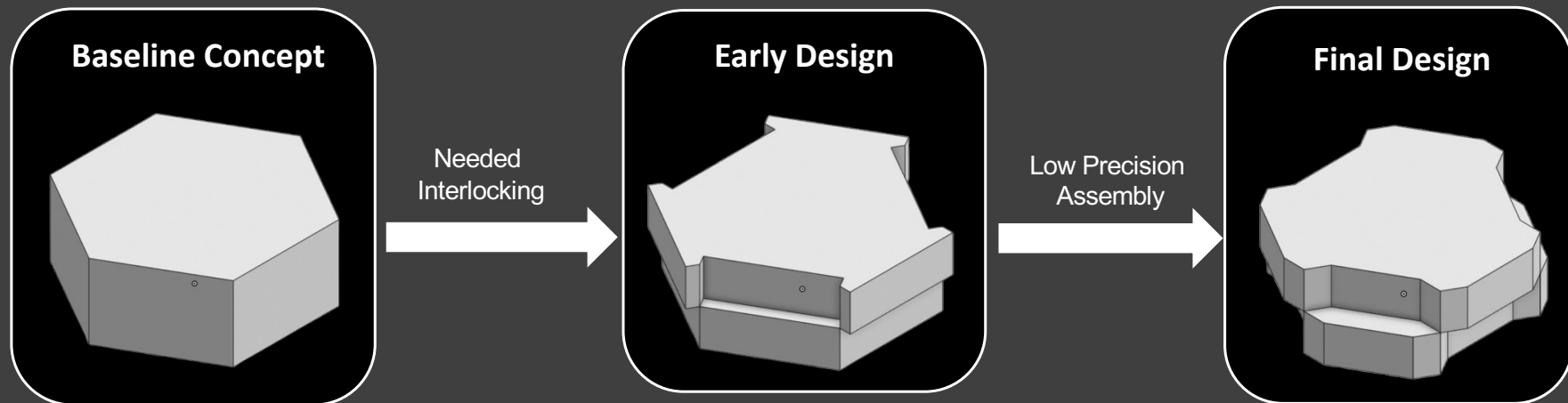
- The tiles will also have to withstand these forces in various failure modes
- Punching shear, bending, cracking, and crushing failure modes are maximized by the landing plumes downward force
- Ablation potentiates through excessive heating
- Ejection potentiates from plume gasses entering between tiles
- Ejection is the most dangerous as ejected tiles can harm the lander and nearby infrastructure



Tile Design

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- **Volume:** Tile dimensions are constrained to the payload bay envelope while minimizing thermal mass to reduce microwave power demand and cycle time.
- **Tileable:** Tiles must tessellate without gaps so sintered output can be assembled into a continuous, functional surface such as a landing pad or habitat floor.
- **Exhaust proofing:** Sintered tiles must achieve sufficient density and surface hardness to resist spalling, cracking, and lofting under repeated lander engine plume impingement.
- **Low precision construction:** Tile geometry must tolerate small rover placement errors on uneven terrain so a functional surface can be assembled without sub-millimeter positioning accuracy.



Agenda

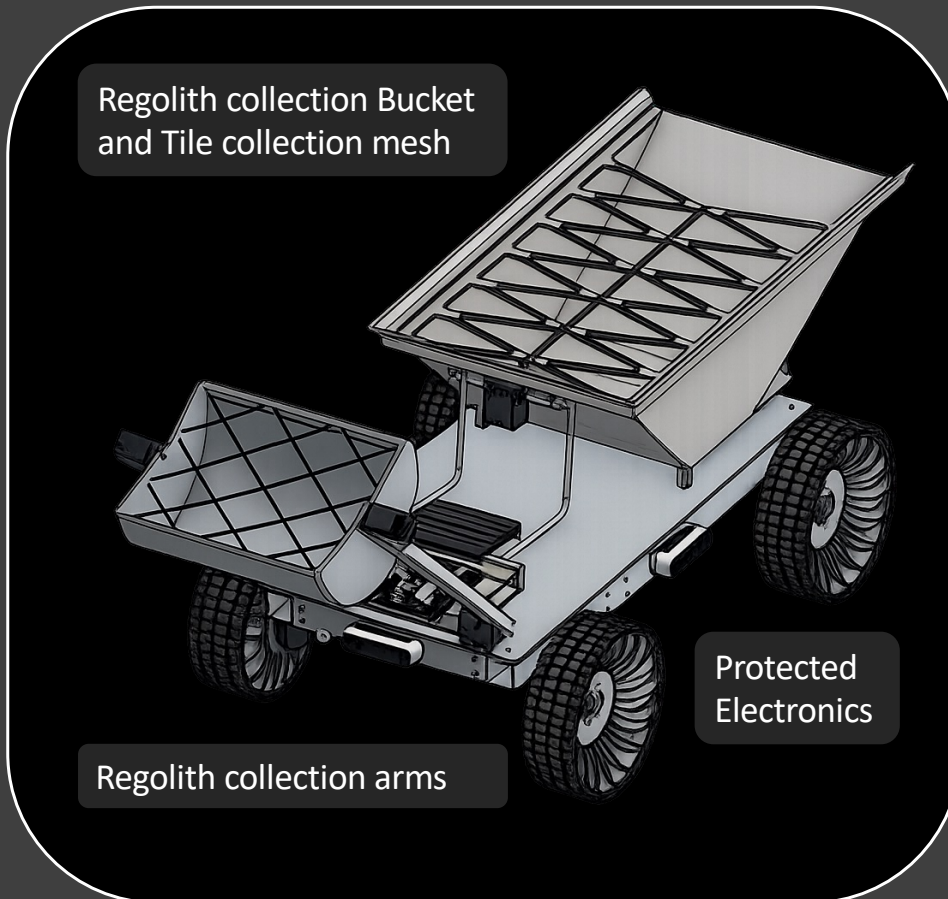
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Rover Diagram

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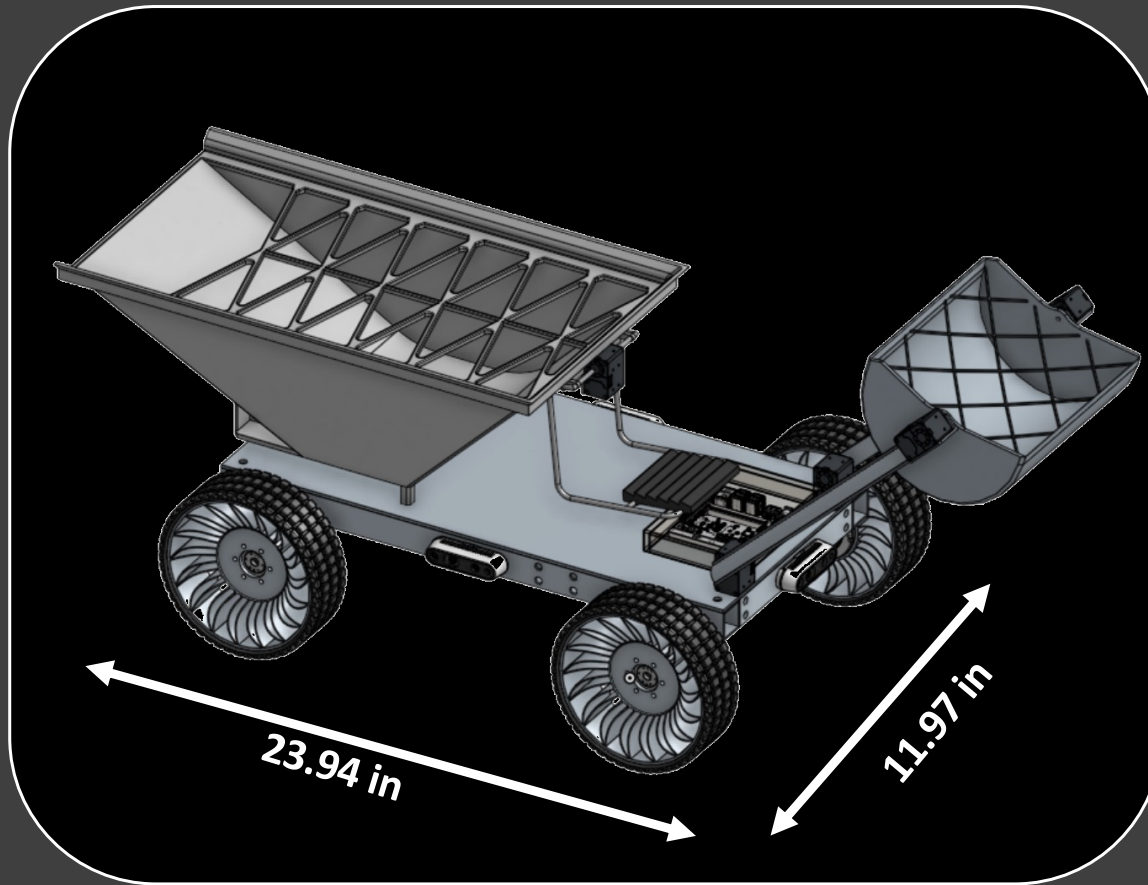


Autonomous Regolith Collection & Delivery Process

- Depart from Griffin lander to begin regolith collection mission
- Navigate to designated regolith collection site
- Collect regolith until payload mass reaches 16 kg
- Return to tile-making station with collected material
- Deposit regolith into processing system
- Await command:
 - Return to charging station **or**
 - Repeat collection cycle from tile-making location
- Collect finished tile and place it at designated location

Mechanical (Bucket/Arm/Scoop)

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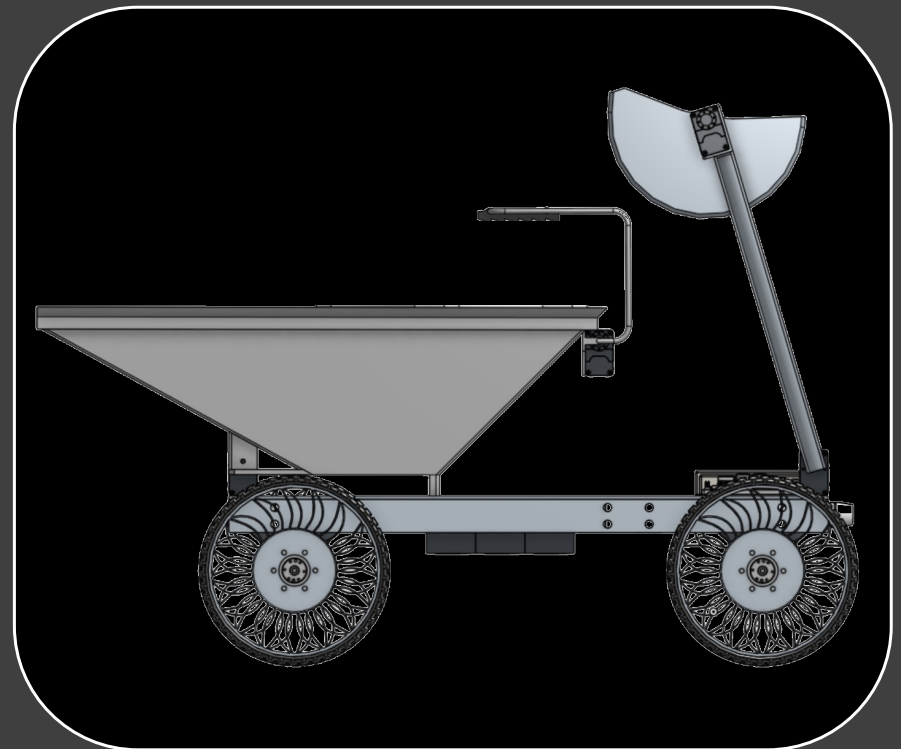
Lunar Rover

- Lightweight (20 kg) rover designed for short-range lunar operations
- Equipped with a 2-DOF robotic arm and integrated hopper for material handling
- Capable of collecting regolith at a rate of 1 kg/min
- Onboard storage capacity of 5.65 liters of regolith

Rover Subsystem Overview

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- **Avionics & Control:** Rad-hard Cortex-M7 MCUs manage autonomy, motors, and safety.
- **Mobility & Actuation:** Space-rated maxon motors for arm; CubeMars motors for drive (require major space modifications).
- **Navigation & Sensing:** LN-200S IMU + stereo cameras with EKF/visual odometry
- **Power System:** 6S Li-ion battery (~400 Wh, 2.8 kg) supports mixed voltage loads (3–60 V across subsystems).
- **Thermal Control:** Kapton heater maintains electronics temperature → lightweight but requires space qualification.



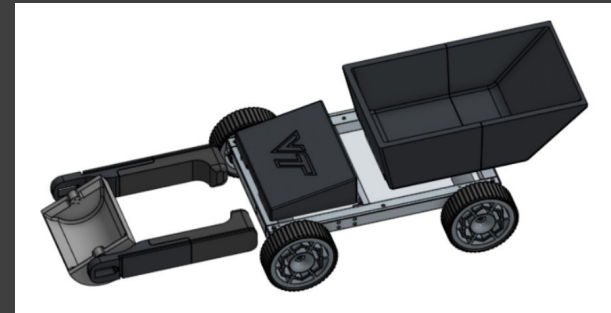
Rover Prototype

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Rover Prototype Details:

- 18 in x 11 in.
- Aluminum piping and plexiglass chassis.
- 3D printed scoop, arms, bucket, and wheels.
- All structural components are mounted with rivets.

Runs	Run Time (s)	Sand Collected (kg)	Collection Rate (kg/min)
Run 1	239	3.460	0.87
Run 2	238	4.300	1.08
Run 3	236	4.045	1.03
Run 4	234	4.745	1.21
Run 5	240	4.285	1.07
Average	237.6	4.167	1.052



Rover Prototype - Electronics

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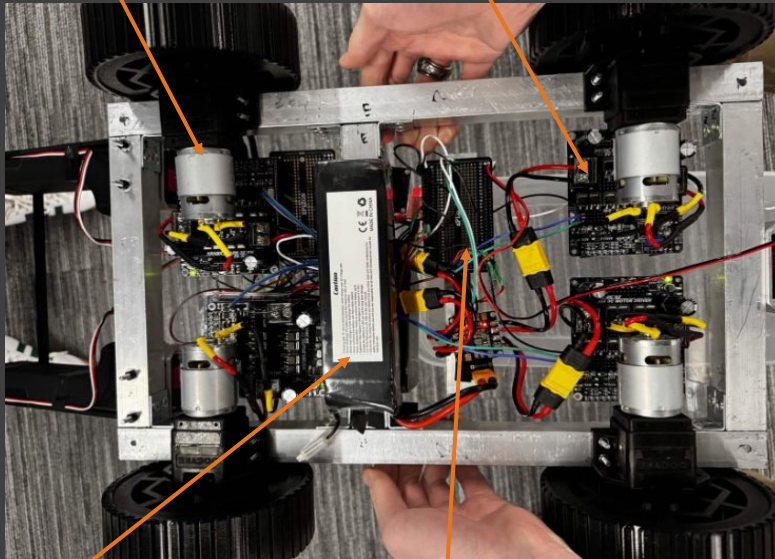
4x 12 V Motors

4x Motor Controllers

2x 25 kg cm Servo Motors
(Not Pictured)

3x 125 kg cm Servo Motors

2200 mAh Lipo Battery



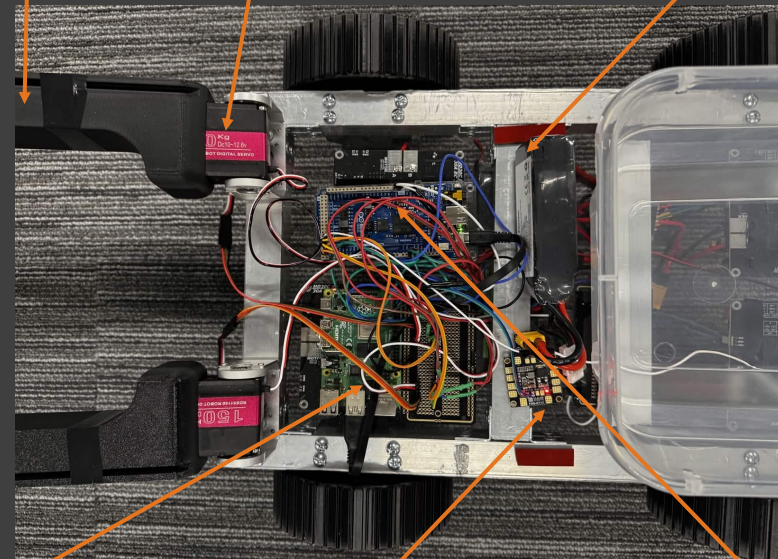
6700 mAh Lipo Battery

2x Power Distribution Bus

Raspberry Pi

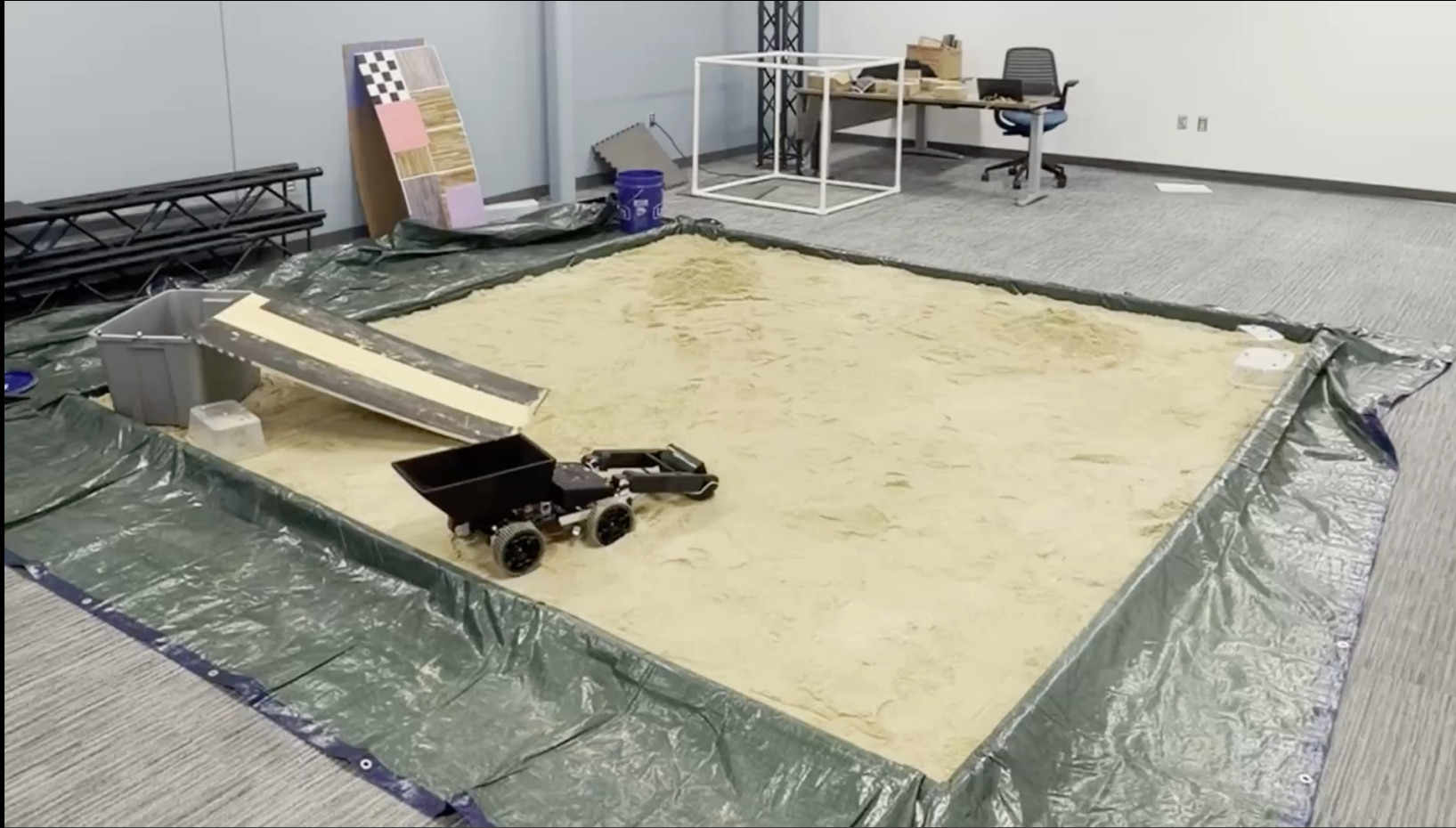
2x Power Distribution Boards

Arduino



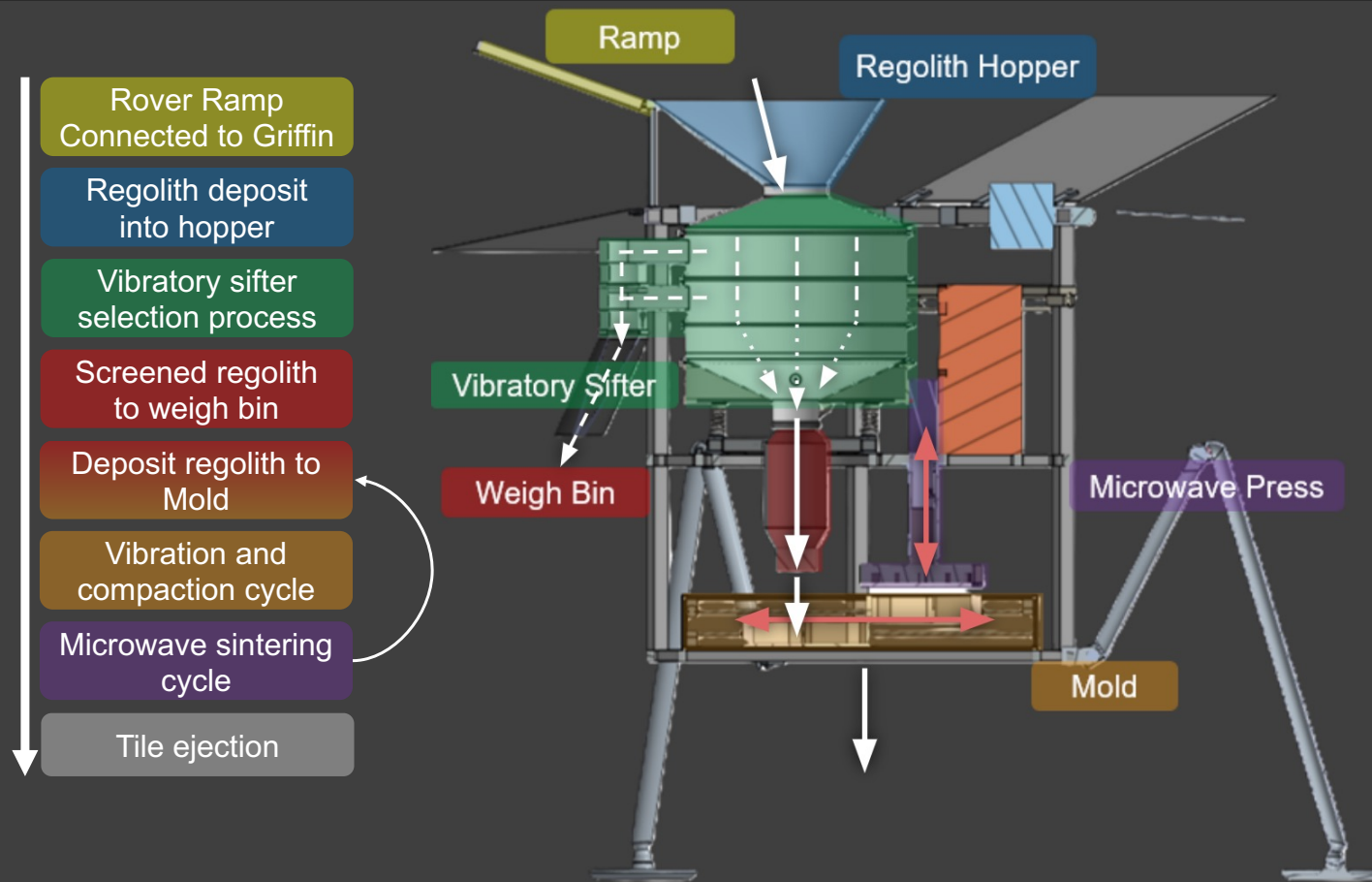
Prototype Video

PROJECT HESTIA



STiM Architecture

PROJECT HESTIA



Maximum structural G Loading, $n = 2$: 24 G

Typical Launch: 3-6 G

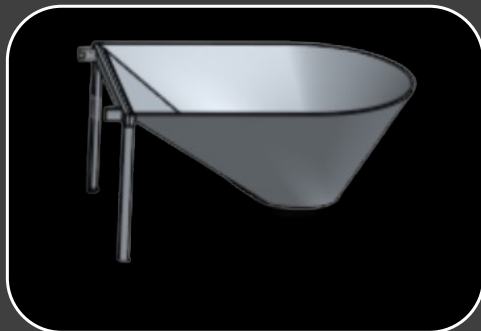
Volume within constraints: 1.573 cubic meters

Filtration

PROJECT HESTIA

Hopper

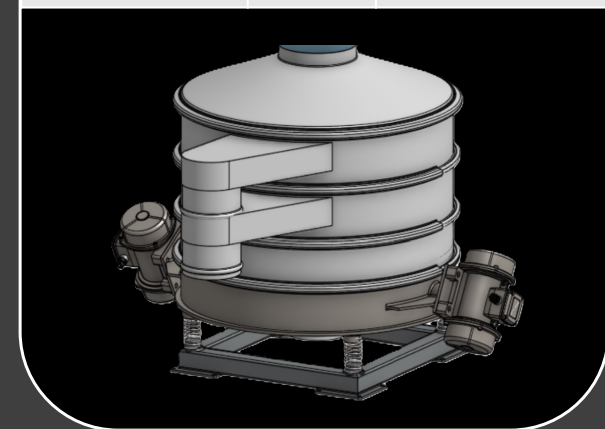
- Rover guided via ramp
- Regolith deposit
- Regolith accumulation prevention:
 - Graded walls (45°)
 - Vibration sequence
- 4 mm thick 6061-T6 aluminum walls
- Capable of holding ~4.5 tiles' worth of regolith (0.026 m³)



Vibratory Screener

- Mates to reception hopper outlet
- Optimal granularity
- Oversized chunks sent to chute
- Stage 1: Removal of oversized rocks and clasts, durable laser cut plate
- Stage 2: Secondary reject, too coarse for sintering, laser cut plate protects fine mesh
- Stage 3: Final cut, fine mesh
- 2x 25 W 24V Brushless DC Motors

Parameter	Unit	Value
Cut Size	d _{max}	600 μm
Mesh Layer 1	d _{m1}	~2.0 mm
Mesh Layer 2	d _{m2}	~0.8 mm
Mesh Layer 3	d _{m3}	500 μm
Feed Rate	ṁ	15 kg/h
Deck Size	D	0.522 m



SYS2.0

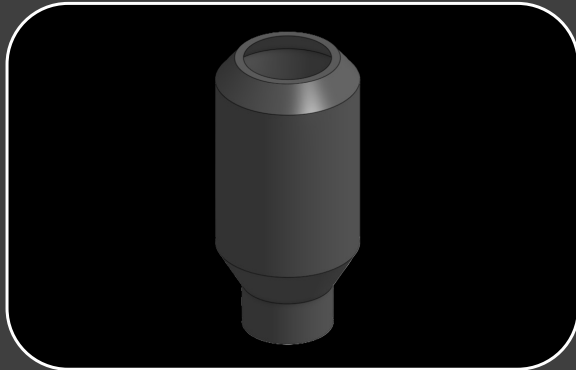
The system shall accept and process lunar regolith to use as the primary material for tile production.

Deposit / Mold

PROJECT HESTIA

Weigh Bin

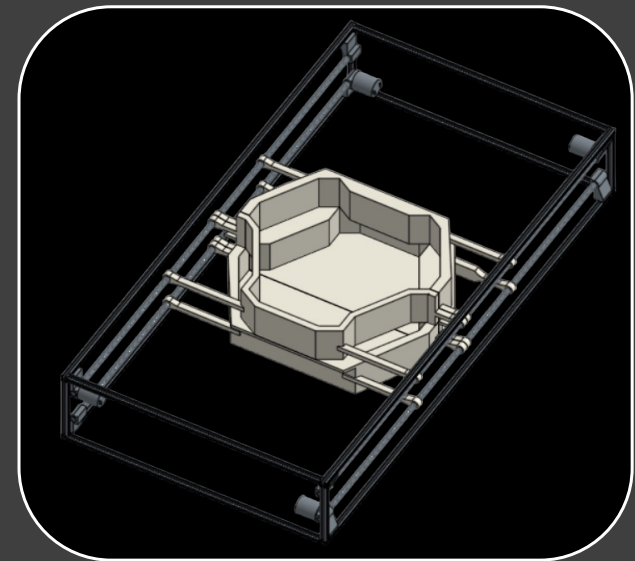
- Tube mating to vibratory screener
- Bin supported by compression load cells
- Inlet gate halts flow from screener when $m_{target} = 12.8\text{ kg}$ is reached
- Outlet rotary gate deposits $m_{layer} = 0.476\text{ kg}$ for each layer



Sinter Mold

- Controlled mass dump into sintering mold
- 2 part mold
- 4x mounted Maxon APES linear actuators
- Vibration and compaction cycle commences
- Compacting plate working in junction with vibrating mold to reach requisite bulk density of $\rho_b = 2.14\text{ g/cm}^3$

Parameters	Unit	Value
Vibration Frequency	f	50 hz
Amplitude	A_{pp}	1 mm
Compaction	σ_c	$\sim 0.4\text{ Mpa}$



Microwave Press

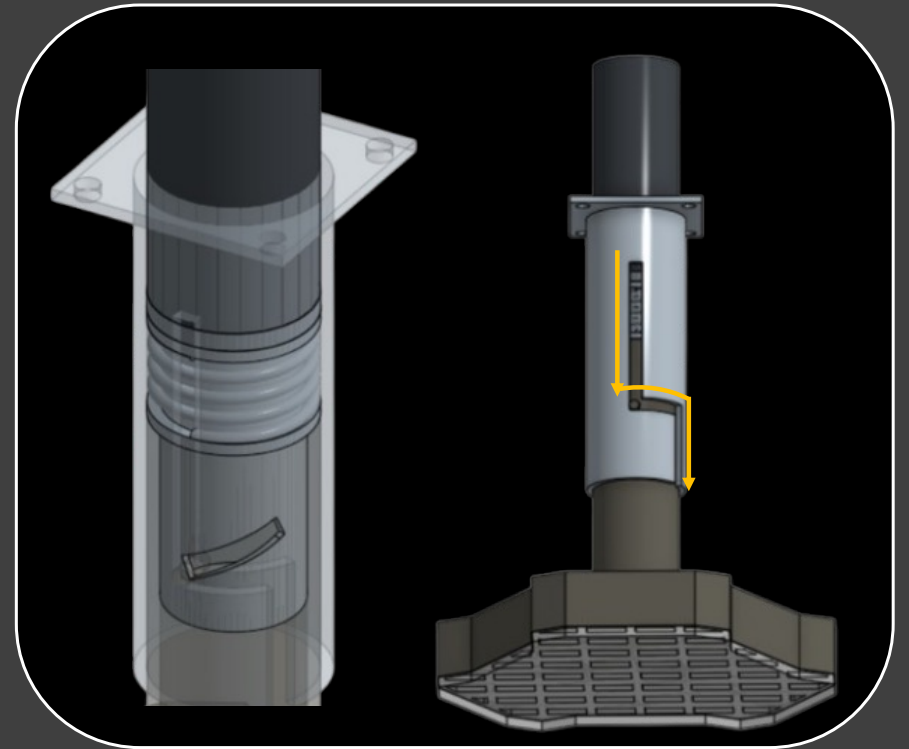
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Microwave Sintering Head

- Tile shaped custom waveguide 915 MHz antenna
- Quartz head is microwave transparent
- Enables full control of field strength
- Even heating

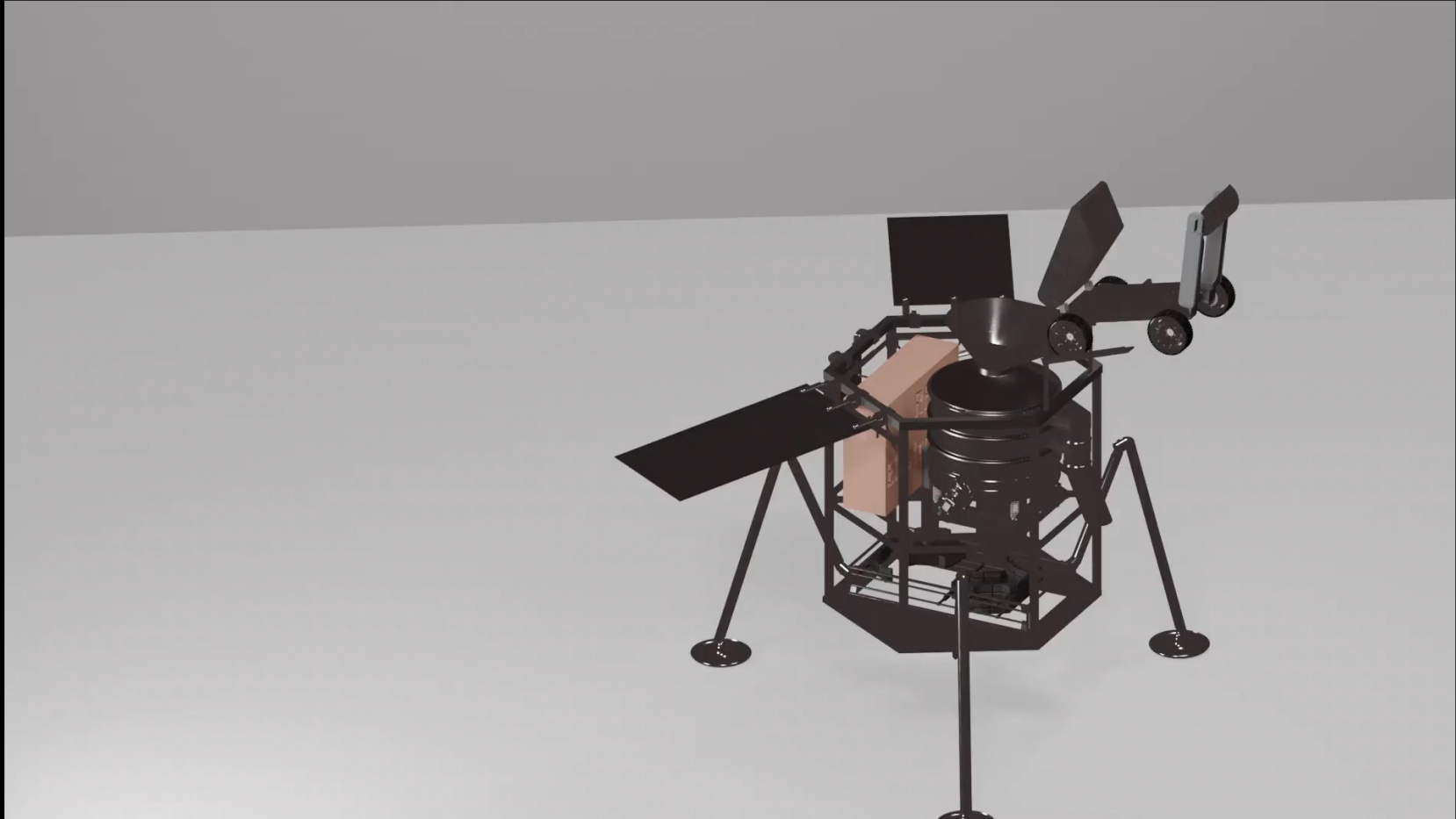
Press Actuation

- Single axis actuator
- Spring and bearing control motion
- Slot controls path
- Tuned to 30° and 53mm



SYS1.0

The system shall autonomously manufacture structurally sound tiles using In-Situ Resource Utilization (ISRU)



Thermal Cooling Trade Study

PROJECT HESTIA

The Thermal Challenge: STiM must reject a massive 1.64 kW heat load in a lunar vacuum where standard convective cooling is difficult.

The Winning Architecture: Passive Heat Pipes provide zero-power thermal transport, while motorized hinges allow the radiators to stow at night and physically shake off hazardous lunar dust.

Heat Pipes + Radiators	Mechanically Pumped Fluid Loop	Phase Change Material (PCM)	Pure Conduction (Solid Metal)
Zero pumping power	Flexible heat routing	Excellent transient buffering	Zero moving parts
Hinge clears lunar dust	Highly scalable (>10 kW)	Extends peak runtime	Low manufacturing cost
Rigid routing constraints	Single-point failure (pump)	Requires massive wax weight	Severe thermal hot-spots
Micrometeoroid strike risk	Heavy plumbing mass	Cannot reject net heat	Extreme mass penalty

Technology	Mass (kg)	Power Draw (W)	Complexity Score (1-10)	Dust Mitigation Score (1-10)	Results
Weights	0.40	0.30	0.15	0.15	1.00
Mechanically Pumped Fluid Loop	~35.00	~50 W	8	1	0.16
Pure Conduction to Walls	> 65.00	0 W	1	1	0.17
Phase Change Material (PCM)	~45.00	0 W	2	1	0.23
Heat Pipes + Radiators	17.31	~5 W (Intermittent)	3	9	0.44

Note: Raw literature values are shown where available; qualitative criteria were converted to scores for the weighted total



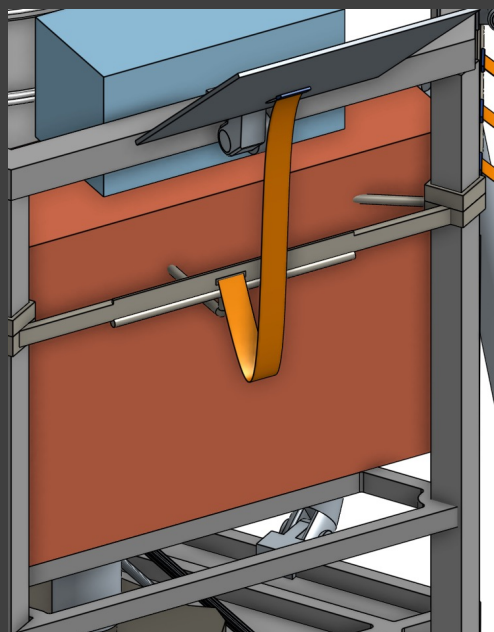
References: [32-33]

STiM Cooling System

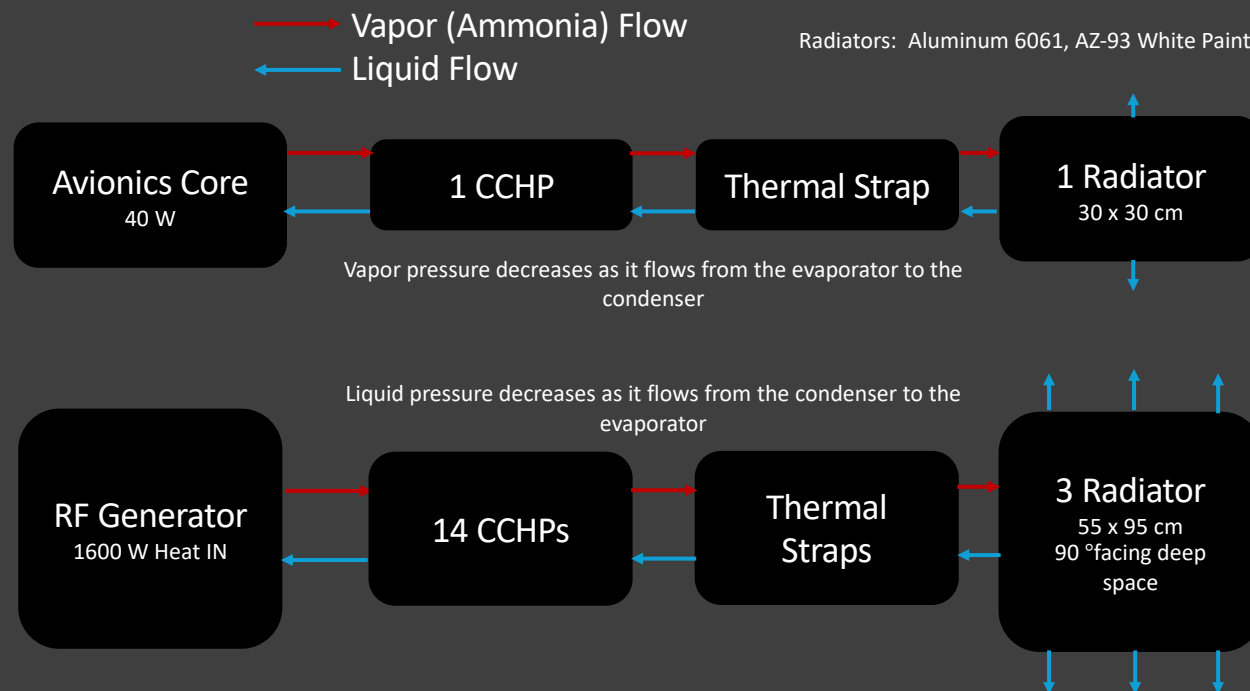
PROJECT HESTIA

2-Loop Hybrid Architecture:

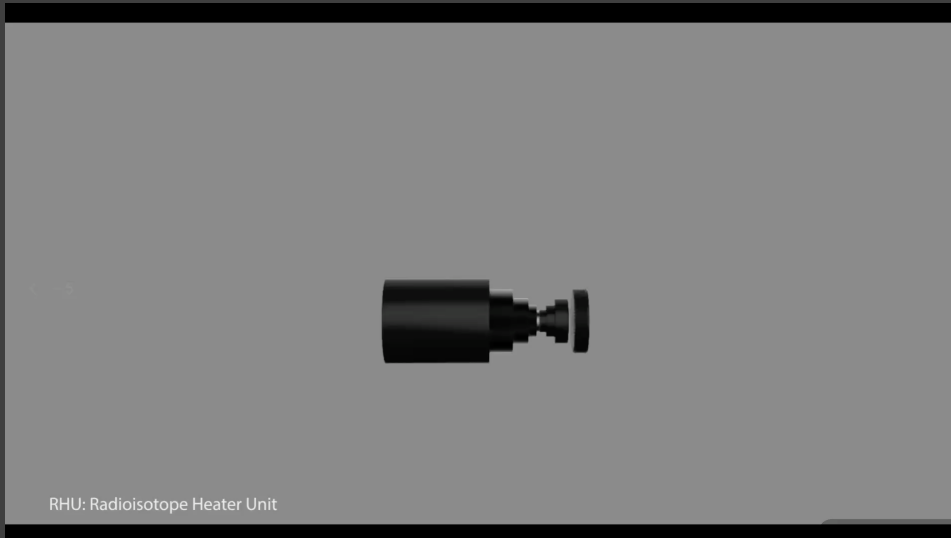
Physically separates sensitive Avionics from the massive RF Generator heat loads.



Expanded View for Cooling System

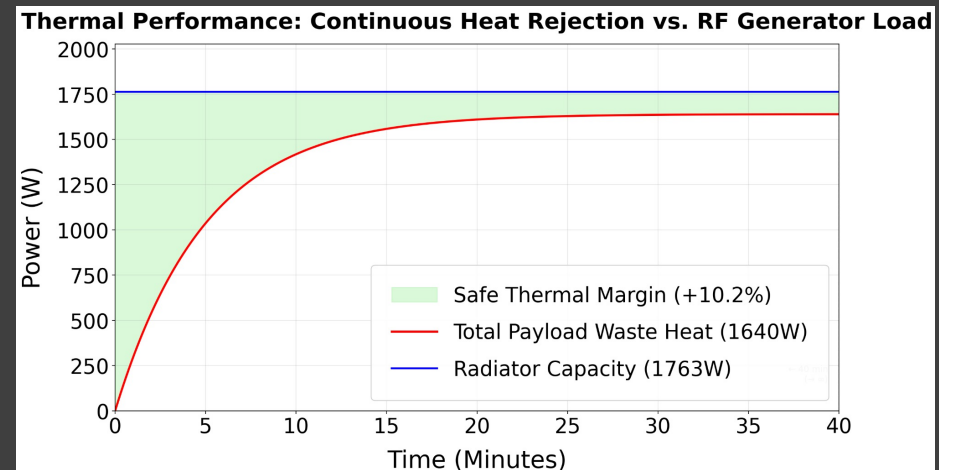


Capillary Liquid Return (Coil): Once the vapor condenses back into liquid at the radiator, the internal grooved wick (the capillary structure) physically pulls the liquid ammonia back to the RF Generator in a continuous, zero-power loop



NASA's RHU: Radio Heater Unit

Zero-Power Night Survival: Motorized Avionics panel folds shut at dusk. Using just two NASA RHUs, the core effortlessly survives the 1.6 W parasitic leak during the 3–4-day lunar deep freeze at the South Pole.

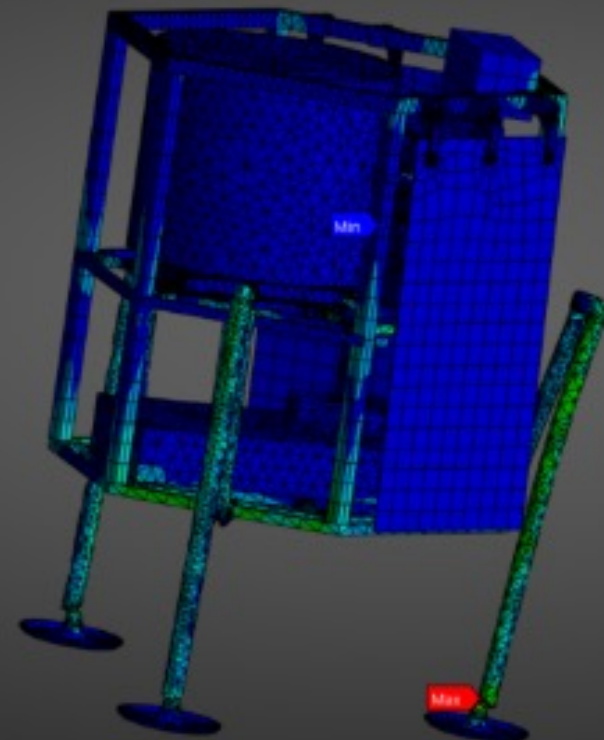
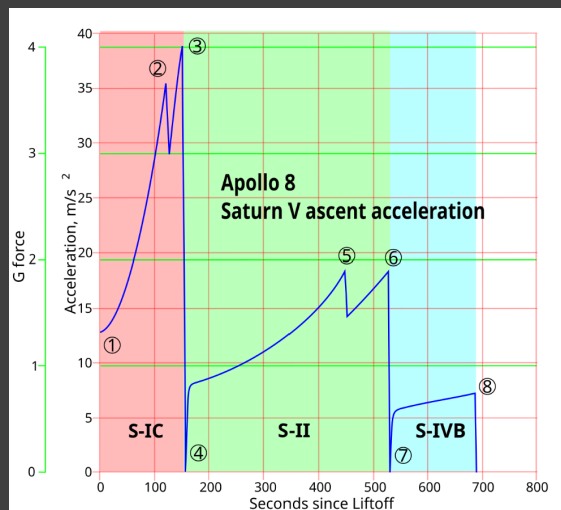


Thermal Performance Graph: Continuous Sintering: three large passive panels provide 1,763 W of constant cooling.

Structural Analysis and Material Selection

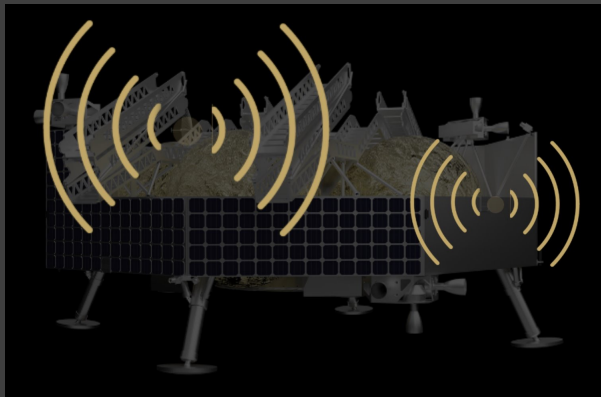
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- Structural analysis done at an acceleration of 10 g
- STiM can safely survive accelerations of 9 g or greater at launch
 - A typical Lunar mission does not exceed 6 g unless aborting



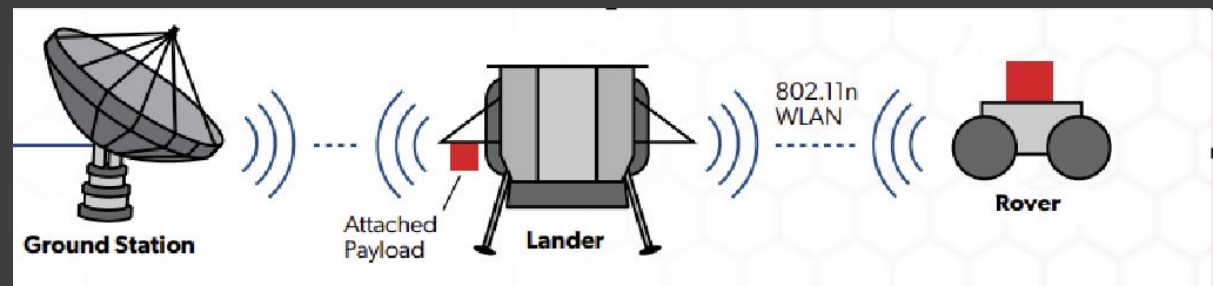
COMMUNICATION SYSTEM:

- The Griffin Lander provides the main relay to Earth through a high-powered transponder and a high-gain antenna.
- Rover uses a 2.4 GHz Wireless Local Area Network modem, compliant with IEEE 802.11n.
- The Rover communicates through the lander using low-gain antenna



SYSTEM REQUIREMENTS:

- Rover sends health, status, and payload telemetry to the lander
- Near-real-time telemetry with low-to-moderate data rate is sufficient
- Full-time observer is not required; only operator commands for supervision and updates.
- Communication must work over the full rover operating range



Agenda

PROJECT HESTIA

1. Team Introduction
2. Project Introduction
3. Project Overview
4. Regolith Analysis
5. Architecture Analysis
6. Conclusion



Testing

PROJECT HESTIA

On-site Testing

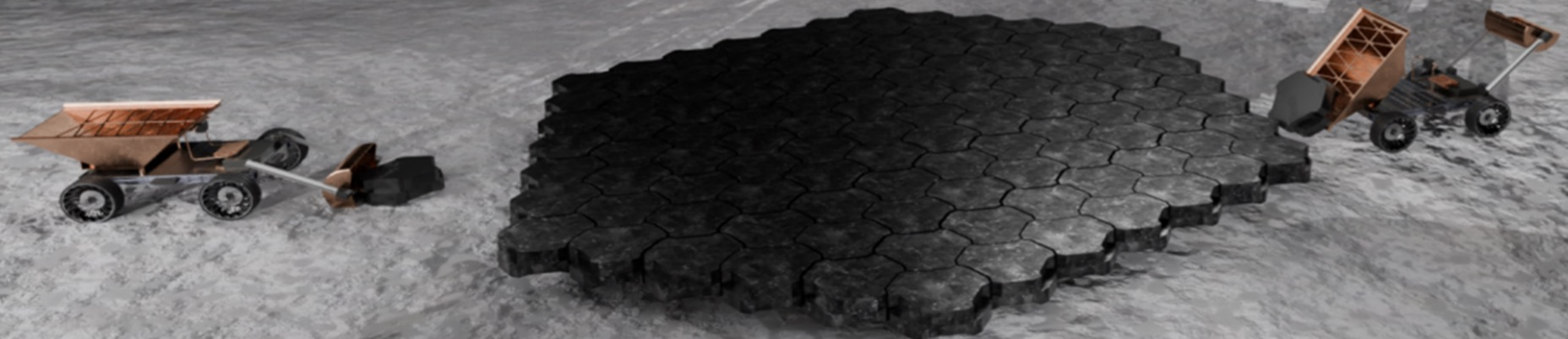
- Thermal Test under Griffin Landers Thrusters
- Stability of interlocking design confirmation
- Alpha particle spectrometer

On Earth Testing

- Sample tiles collected and returned to Earth by astronauts for further testing
- More in depth tests such as strength and impact tests will be conducted

Advanced Testing Technology (Future)

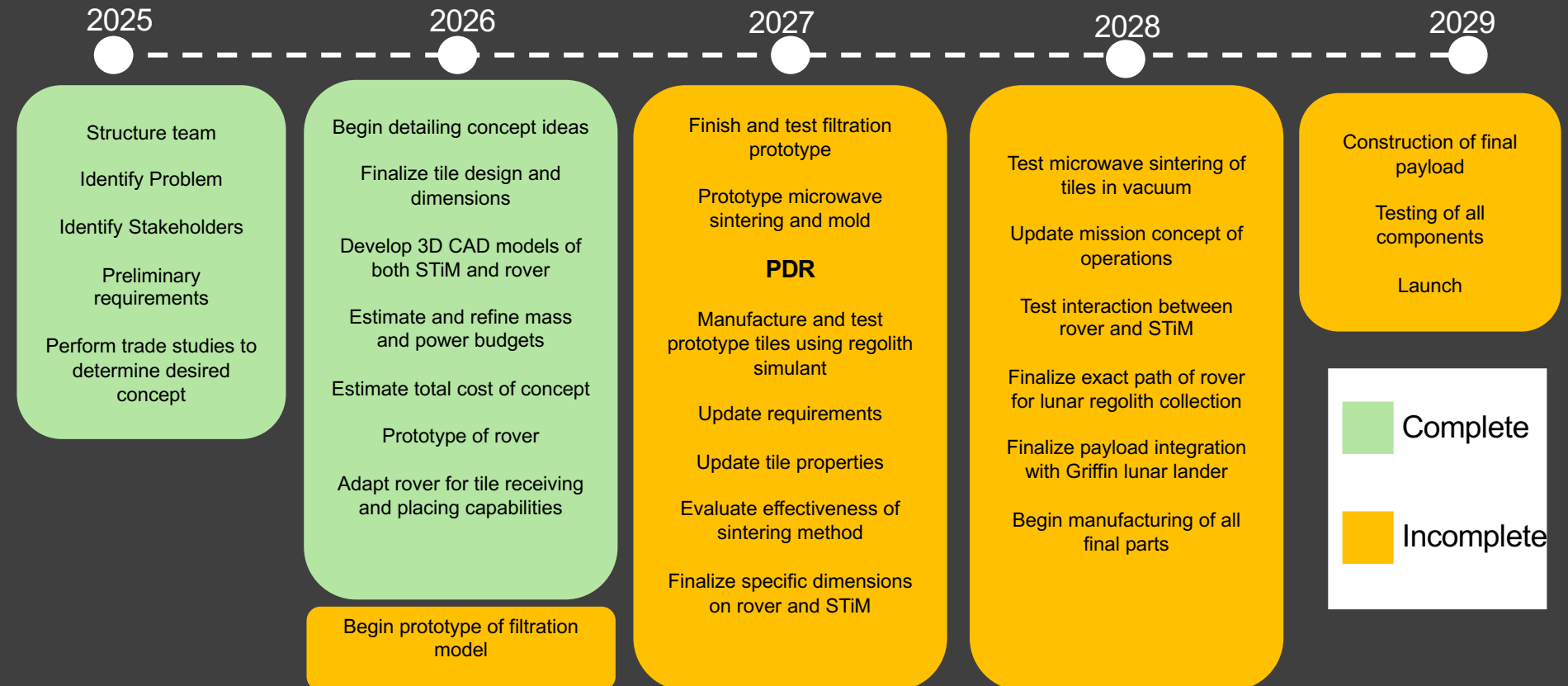
- Ultrasonic penetrating laser to scan for imperfections in the tile



SYS3.0 The system shall produce tiles that meet defined structural and thermal performance standards.

Path to PDR

PROJECT HESTIA



Mission Risk Analysis

Class C: Moderate risk tolerance mission

PROJECT HESTIA

Risk ID	Risk Statement	Consequence	Mitigation Strategy								
				Consequence of Failure	5	4	3	2	1	Likelihood of Failure	Risk Classification
M1	The microwave sintering system fails to operate as expected once deployed on the lunar surface	The tiles may be weak or poorly formed, causing them to crack or break during placement or lander operations.	Test the system on Earth using lunar regolith simulant and make the necessary adjustments before launch.								R (Rover)
M2	Solar panels dust accumulation	Reduced available power for rover and tile maker operations	Controlled dumping of regolith and dust shield								M (Mission)
P1	Microwave sintering efficiency is lower than expected	More energy is required, reducing tile production performance	Validate sintering efficiency through extensive testing beforehand								E (Environmental)
P2	Available power is insufficient for sustained sintering operations	Incomplete processing and reduced tile output	Budget power conservatively								P (Power and Thermal)
R1	Rover is unable to traverse terrain and gets stuck	Regolith delivery to the STiM is interrupted	Flat site selection and limited collection radius								St (Structural)
E1	Landing site regolith is unsuitable for consistent tile production	Reduced sintering quality, leading to increase in production time	Select a location with stable and predictable regolith								
E2	Sintering system overheats during operation	Processing must stop or hardware may be damaged	Implementing a cooling system								
St1	Produced tiles do not meet the required structural quality for construction use	Manufactured tiles cannot be used for the intended infrastructure application	Include demonstration tile production as a secondary success criterion								

Rover Budgets

PROJECT HESTIA

Rover Mass Budget					Rover Cost Budget		Rover Power Budget		
Component:	Mass (kg):	GA	MGA (kg)	Total MGA (kg)	Category:	Cost (USD M):	Mode:	Power Draw:	Description
Battery Pack	3.56	5%	0.178	3.74	Power & Thermal	10.8	Controls	<200 W	Driving, Dumping, Comms.
Electronics Components	3.99	10%	0.399	4.39	Navigation Sensors (IMU + Cameras)	5.30	Digging/Scoop/Dump	<175 W	Collecting Regolith.
Chassis	3.25	2%	0.0650	3.31	Structure & Mechanical Systems	12.7	Total Battery:	480 Wh	Enough power for multiple trips a day.
Wheels + Motors	4.00	2%	0.0800	4.08	Electronics & Avionics	8.00			
Tires	0.960	5%	0.0480	1.01	Actuation (Motor & Gearboxes)	3.30			
Bucket	3.27	15%	0.491	3.77	Software & Integration	15.3			
Scoop + Lift Arms	1.57	10%	0.157	1.73	Labor Cost	10.0			
Harness	0.820	5%	0.0410	0.86	Testing Cost	18.0			
Total Mass	21.4	7%	1.50	22.9	Rover Total Cost:	83.4			



STiM Budget

PROJECT HESTIA

STiM Mass Budget					STiM Cost Budget		STiM Power Budget		
Component:	Mass (kg):	GA (%)	MGA (kg)	Total MGA (kg)	Category:	Cost (USD M)	Mode:	Power Draw:	Description
Aluminum Frame	38.9	2%	0.778	39.7	Structure & Mechanical Systems	65.6	Sintering	4.50 kW	1 kW RF, Press, Controls
Support Legs	13.5	5%	0.675	14.2	Electronics & Avionics	14.2	Ejecting Tiles	0.40 kW	Move finished bricks off the STiM
Battery Pack	30.0	5%	1.50	31.5	Actuation (Motor & Gearboxes)	10.0	Conditioning	250 W	Vibratory methods, Release methods
Vibratory Motors	6.35	5%	0.32	6.67	Regolith Processing (Microwave & Mold)	31.7	Idle	Day: 50 W Night: <10 W	Day – electronics ready, Night – survival, heat, and comms
Regolith Processing System	16.6	15%	2.49	19.1	Power & Thermal	51.4			
Cooling System	17.3	10%	1.73	19.0	Software & Integration	32.5			
Hopper & Ramp	5.47	5%	0.274	5.74	Labor Cost	20.0			
Mold and Sintering System	20.4	10%	2.04	22.4	Testing Cost:	45.0			
Environmental / Radiation Shielding	17.0	5%	0.850	17.9	STiM Total Cost:	270			
Harness	8.28	5%	0.41	8.69					
Total Mass:	174	6%	10.4	184					
Total System Mass:	194			207	Total System Cost:	~353			



Project HESTIA Conclusion

PROJECT HESTIA

Power Draw:
5 kW

Structures can
withstand over 10 G's

Mission Timeline: Six-month window
between September 2029 and March 2030

Dimensions:
STiM - 1.1m x 1.1m x 1.3m (L x W x H)
Rover - 1.2m x 0.5m x 0.4m (L x W x H)

Estimated Cost:
\$353 M

Tile Production Time:
7.32 hours

Total Payload Mass:
194.4 kg





QUESTIONS?

- [1] <https://cosmicspace.org/2025/07/2025-26-cosmic-capstone-challenge-c3-announcement/>
- [2] <https://www.nasa.gov/humans-in-space/commercial-space/the-perils-and-promise-of-dust-on-the-moon/>
- [3] https://ntrs.nasa.gov/api/citations/20250011152/downloads/Lunar%20Landing%20Pad_2025-12-06.pdf
- [4] https://www.astrobotic.com/wp-content/uploads/2022/01/PUGLanders_011222.pdf
- [5] Slide 17 - No stable link yet (Hyunwoo Jin et al., KLS-1 sintering, 2024)
- [6] <https://pdf.sciencedirectassets.com/312002/1-s2.0-S2352710224X00054/1-s2.0-S2352710224007617/main.pdf> [EXPIRES - find DOI]
- [7] <https://pdf.sciencedirectassets.com/271630/1-s2.0-S0955221924X00163/1-s2.0-S0955221924009208/main.pdf> [EXPIRES - find DOI]
- [8] <https://pdf.sciencedirectassets.com/271475/1-s2.0-S0950061824X00410/1-s2.0-S0950061824042740/main.pdf> [EXPIRES - find DOI]
- [9] Slide 17 - No stable link yet (sintering lunar regolith simulant domestic microwave, 2024)
- [10] https://ipnpr.jpl.nasa.gov/progress_report/42-176/176C.pdf
- [11] <https://www.nasa.gov/news-release/nasa-identifies-candidate-regions-for-landing-next-americans-on-moon/>
- [12] <https://www.nasa.gov/news-release/nasa-provides-update-on-artemis-iii-moon-landing-regions/>
- [13] <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JE009127>
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- [17] <https://www.nasa.gov/news-release/nasa-provides-update-on-artemis-iii-moon-landing-regions/>
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- [22] <https://www.geeksforgeeks.org/electronics-engineering/microstrip-patch-antenna/>
- [23] <https://articles.adsabs.harvard.edu/full/1973LPSC....4.3133O/0003133.000.html>
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- [25] <https://www.sciencedirect.com/science/article/pii/S0094576521003349>
- [26] <https://articles.adsabs.harvard.edu/full/1972LPSC....3.3157B/0003157.000.html>
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- [39] <https://www.sshlmachinery.com/how-much-force-can-a-hydraulic-press-produce.html>
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BACKUP SLIDES

In Depth StiM Cooling Budget

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Cooling Loop	Component Breakdown	Mass (kg)
Loop 1: Avionics	Avionics Radiator Panel (30x30 cm)	1.36
	Constant Conductance Heat Pipe (1x) & Flex Strap	0.05
	Motorized Hinge & NASA RHU Pellets (2x)	0.48
	Avionics Subtotal:	1.89
Loop 2: Microwave	Large Radiator Panels (3x, 55x95 cm)	12.26
	Constant Conductance Heat Pipes (14x) & Flex Straps	1.06
	Motorized Hinges (3x, 12V DC Gearmotors)	2.10
	Microwave Subtotal:	15.48
TOTAL	FULL THERMAL SYSTEM MASS:	17.31

Cooling Calculations

PROJECT HESTIA

$$Q = \varepsilon \cdot \sigma \cdot A_{\text{eff}} \cdot T^4$$

Given Variables (White Paint)

$$\sigma = 5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$$

$$\varepsilon = 0.91$$

$$T = 333 \text{ K}$$

Effective Area Calculation

Area of ONE full panel:

$$0.55 \times 0.95 = 0.5225 \text{ m}^2$$

Panel 2 Full x 1 cutout (leg clearance):

$$0.60 \times 0.11 = 0.066 \text{ m}^2$$

$$0.5225 - 0.066 = 0.4565 \text{ m}^2$$

Total for 3 panels (2 full + 1 cutout):

$$(2 \times 0.5225) + 0.4565 = 1.5015 \text{ m}^2$$

Both sides radiating:

$$A_{\text{eff}} = 1.5015 \times 1.85 = 2.778 \text{ m}^2$$

Final Heat Rejection Calculation

$$Q = (0.91)(5.67 \times 10^{-8})(2.778)(333^4)$$

Step-by-step:

$$0.91 \times 5.67 \times 10^{-8} = 5.1597 \times 10^{-8}$$

$$333^4 = 12,296,370,321$$

$$(5.1597 \times 10^{-8}) \times 2.778 = 1.4334 \times 10^{-7}$$

$$Q = (1.4334 \times 10^{-7}) \times (12,296,370,321)$$

Final Answer:

$$Q \approx 1,763 \text{ W}$$

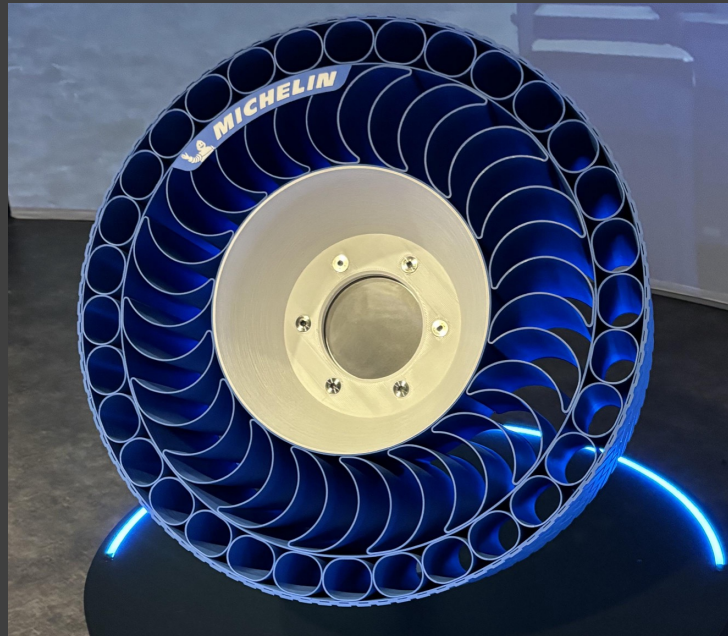
Margin Check

$$\text{Heat generated (RF Loop 2)} = 1,600 \text{ W}$$

$$\text{Heat rejected} = 1,763 \text{ W}$$

$$\text{Margin} = (1763 - 1600) / 1600 \approx 0.102 \approx 10.2\%$$





Electronics

PROJECT HESTIA

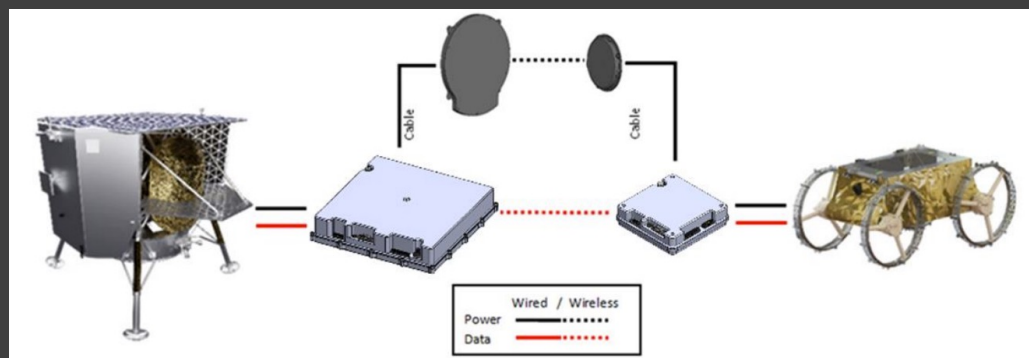
Subsystem / Component	Part Name	Space Grade / Edits Needed	Mass	Size	Power Requirements
Main Autonomy Computer	Rad-Hard 32-bit Arm Cortex-M7 MCU	Space grade	18 g	~28 × 28 × 3–4 mm	3.3 V I/O, 1.8 V core
Wrist Motor	maxon EC 32 flat SPACE + GPX 32 UP (3-stage)	Near-space-grade; minor integration (thermal, harnessing, controller)	~372 g	Ø32 mm; length 66.5 mm	24 V, 0.421 A, ~15 W
Elbow Motor	maxon EC 45 flat SPACE + GPX 42 UP (3-stage)	Near-space-grade; minor integration	~720 g	Ø45 mm motor + Ø42 mm gearhead	12 V, ~50 W
Drive Motor Controller MCU	(Motor Control MCU)	Space grade	~20 g	~30 × 30 × 3–4 mm	5V logic, up to 60V motor supply
Radio / Communications	2.4 GHz WLAN Relay Communication Module + Low-Gain Antenna	Not fully space grade would require major mods	100–250 g estimated	50 × 50 × 20 mm electronics + small low-gain antenna	0.5 W max
Arm & Hopper Controller MCU	SAMRH71 Rad-Hard Cortex-M7	Space grade	18 g	~28 × 28 × 3–4 mm	3.0–3.6 V I/O, ~1.8 V core
Safety / Power MCU	Microchip LX7720 Motor Driver	Space grade	~20 g	~30 × 30 × 3–4 mm	5V logic, up to 60V motor supply
Hopper Tip Motor	CubeMars RI50 Frameless BLDC	NOT space grade; major mods required	180.8 g	Ø54 × 27 mm	12–48 V, ~500 W
Drive Motors (×4)	CubeMars AK60-6 Actuator (wheel motors)	NOT space grade; major mods required	0.65 kg	Ø42.8 × 12.8 mm	24 V, ~50 W each
IMU (Navigation)	LN-200S IMU	Space grade	~0.5–0.6 kg	~10–15 cm module	~8–15 W
Stereo Cameras (×2)	IM200 Space Camera	Near-space-grade; minor integration	118 g total	~50 × 50 × 50 mm each	~1.4 W total
Navigation Software	EKF + Visual Odometry	Space-appropriate (software validation needed)	N/A	N/A	~5–15 W (shared compute)
Main Battery Pack	6S Li-ion Battery Pack (~400 Wh, 22.2V, ~18Ah)	Space grade	2.80 kg	0.053 m ²	(~400 Wh, 22.2V, ~18Ah)
Electronics Box Survival Heater	Polyimide (Kapton) Etched-Foil Heater	NOT space grade; major mods required	~5–15 g	~75 × 75 mm, ~0.25 mm thick	~5–10 W



Rover Wireless Charger

PROJECT HESTIA

- A wireless lunar charging system developed by Astrobotic + WiBotic will be used for the rover.
- It is specifically designed for harsh lunar environment and tested in simulated lunar conditions. Proven to work in extreme temperatures, harsh dust environments, and on uneven terrain – all at an 80-85% total system efficiency
- Astrobotic is already offering the system for commercial purchase in 125w and 400w configurations.



ADVANTAGES

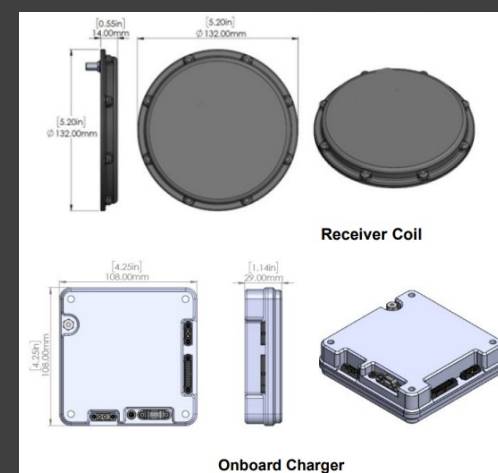
Dust-tolerant charging (tested with simulated regolith)
Alignment tolerance:

≤ 4 cm gap + Up to 40° misalignment

Enables power during lunar night (heater support)

Reduces need for oversized batteries

Lightweight



Budget

PROJECT HESTIA

Rover Mass Budget		Rover Cost Budget		STiM Mass Budget		STiM Cost Budget	
Component:	Mass (kg):	Category:	Cost (USD):	Component:	Mass (kg):	Category:	Cost (USD):
Battery Pack	2.80	Power & Thermal	10,750,000	Aluminum Frame	41.0	Structure & Mechanical Systems	65,625,000
Electronics Components	6.48	Navigation Sensors (IMU + Cameras)	5,250,000	Support Legs	13.5	Electronics & Avionics	14,150,000
Chassis	3.23	Structure & Mechanical Systems	12,650,000	Battery Pack	30.0	Actuation (Motor & Gearboxes)	9,845,000
Wheels	1.09	Electronics & Avionics	8,030,000	Vibratory Motors	6.35	Regolith Processing (Microwave & Mold)	31,700,000
Tires	1.47	Actuation (Motor & Gearboxes)	3,280,000	Regolith Processing System	15.54	Power & Thermal	51,400,000
Bucket	8.16	Software & Integration	15,250,000	Cooling System	17.31	Software & Integration	32,500,000
Scoop + Lift Arms	4.01	Labor Cost	10,000,000	Hopper & Ramp	5.47	Labor Cost	20,000,000
Total Mass	30.24	Rover Total Cost:	65,210,000	Mold and Sintering System	20.39	STiM Total Cost:	225,220,000
				Total Mass	154.9		
				Total Mass:	185.14	Total Cost:	~291,000,000
Rover Power Budget				STiM Power Budget			
Mode:	Power Draw:	Description:		Mode:	Power Draw:	Description:	
Controls	<200 W	Driving, Dumping, Comms		Sintering	4.50 kW	1 kW RF, Press, Controls	
Digging/Scoop/Dump	<175 W	Collecting Regolith		Ejecting Tiles	0.40 kW	Move finished bricks off the STiM	
Total Battery:	480 Wh	Enough power for multiple trips per day		Conditioning	250 W	Vibratory methods, Release methods	
				Idle	Day: 50 W Night: <10 W	Day – electronics ready, Night – survival, heat, and comms	



Stationary Tile Maker Budgets

PROJECT HESTIA

STiM Mass Budget		STiM Cost Budget		STiM Power Budget		
Component:	Mass (kg):	Category:	Cost (USD):	Mode:	Power Draw:	Description
Aluminum Frame	44.34	Structure & Mechanical Systems	65,625,000	Sintering	4.50 kW	1 kW RF, Press, Controls
Support Legs	13.50	Electronics & Avionics	14,150,000	Ejecting Tiles	0.40 kW	Move finished bricks off the STiM
Battery Pack	30.0	Actuation (Motor & Gearboxes)	9,845,000	Conditioning	250 W	Vibratory methods, Release methods
Vibratory Motors	6.35	Regolith Processing (Microwave & Mold)	31,700,000	Idle	Day: 50 W Night: <10 W	Day – electronics ready, Night – survival, heat, and comms
Regolith Processing System	16.60	Power & Thermal	51,400,000			
Cooling System	8.95	Software & Integration	32,500,000			
Hopper & Ramp	5.46	Labor Cost	20,000,000			
Mold and Sintering System	24.90	STiM Total Cost:	225,220,000			
Total Mass	150.1					
Total System Mass:	169.7	Total System Cost:	~291,000,000			

